Causal Analysis Report

BPI_Challenge_2013_open_problems.xes.gz-dfg

Decision points:	2539
Choice sets:	4
Transitions fitted:	17
Significant (p<0.05):	3
Generated:	2026-06-09 15:11

Step 1 — Association test (logistic regression + permutation)
H₀: choice ⊥ util | choice_set.   reject_H0=True → util predicts routing.

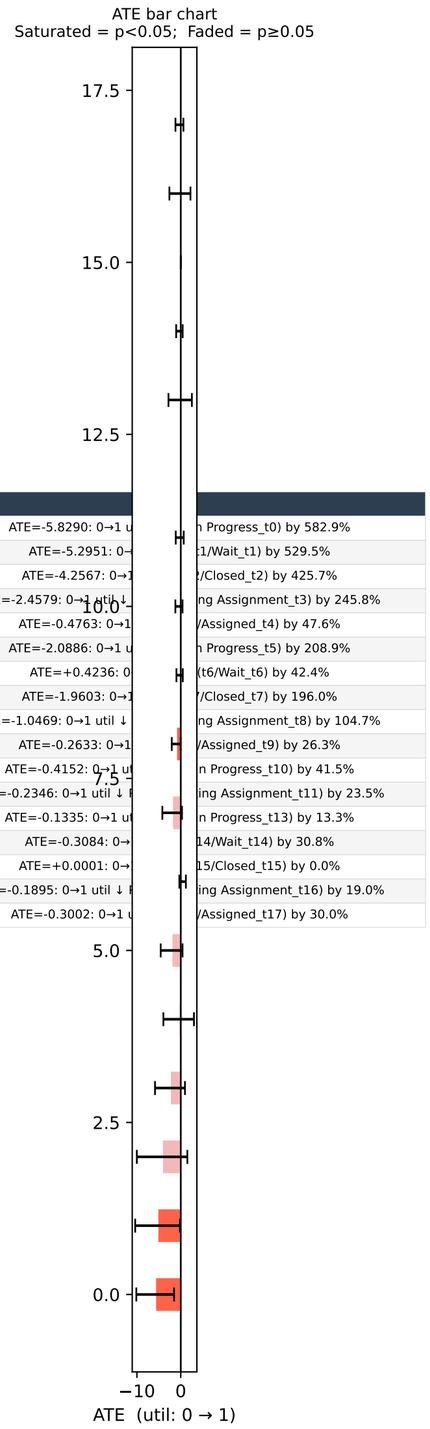
choice_set	choice_set_labels	n	choices_seen	coef	ci_95	perm_p_value	reject_H0	note
(0, 1, 2, 3, 4, 12)	('In Progress_t0', 'Wait_t1', 'Closed_t2', 'Awaiting Assignment_t3', 'Assigned_t4', 'τ_t12')	949	0, 1, 2, 3, 4	0.7428	[0.058, 1.458]	0.0175	True	higher util → t4 (Assigned_t4) over t0 (In Progress_t0)
(5, 6, 7, 8, 9, 12)	('In Progress_t5', 'Wait_t6', 'Closed_t7', 'Awaiting Assignment_t8', 'Assigned_t9', 'τ_t12')	134	5, 6, 7, 8, 9	-1.4078	[-2.372, -0.658]	0.001	True	higher util → t5 (In Progress_t5) over t9 (Assigned_t9)
(10, 11, 12)	('In Progress_t10', 'Awaiting Assignment_t11', 'τ_t12')	273	10, 11	4.9229	[3.763, 5.902]	0.0	True	higher util → t11 (Awaiting Assignment_t11) over t10 (In Progress_t10)
(13, 14, 15, 16, 17)	('In Progress_t13', 'Wait_t14', 'Closed_t15', 'Awaiting Assignment_t16', 'Assigned_t17')	819	13, 14, 15, 16, 17	0.3729	[-0.103, 0.802]	0.156	False	higher util → t17 (Assigned_t17) over t13 (In Progress_t13)

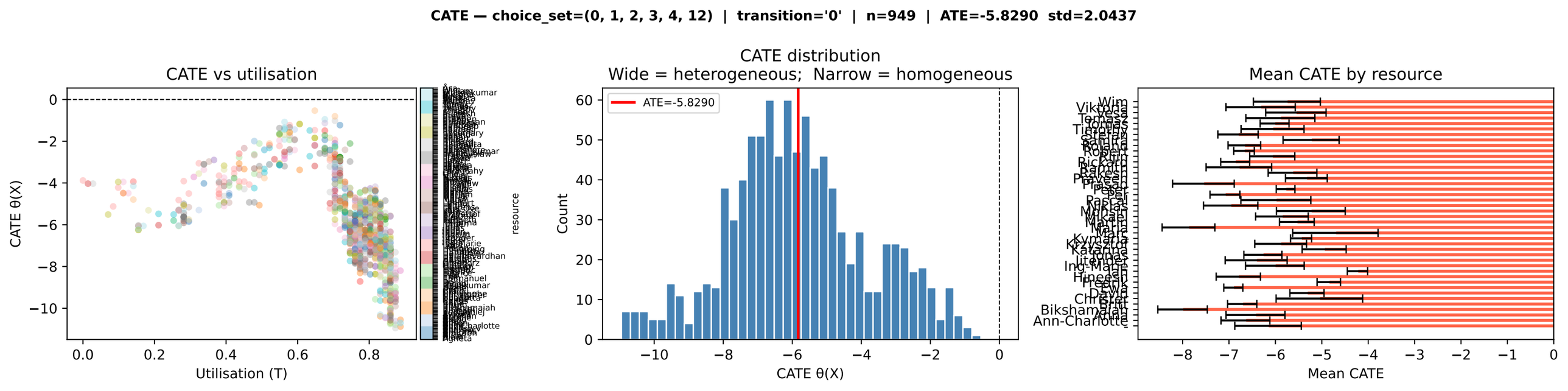
Step 2 — Double ML (partially linear regression, HC3 robust SE)
θ = linear effect of 1-unit ↑ util on P(chose k), after partialling out X.
Equals ATE when the effect is homogeneous. Otherwise: variance-weighted average of CATEs.

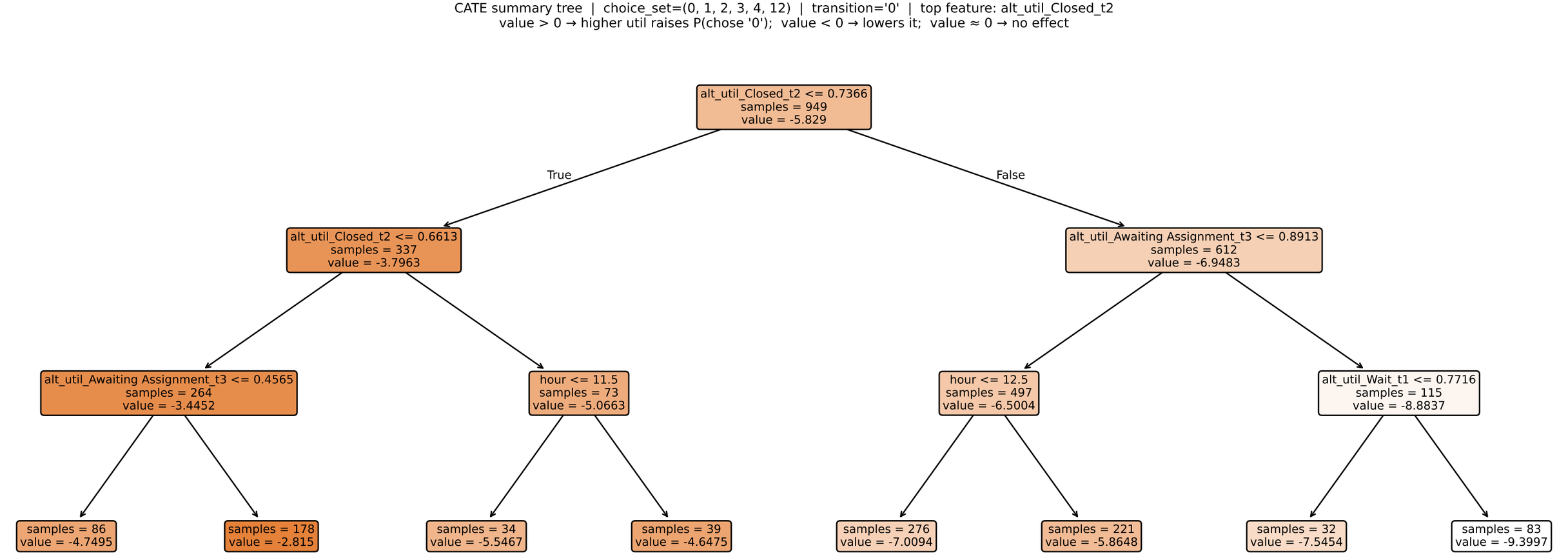
choice_set	choice_set_labels	n	choices_seen	transition	transition_label	alt_utils	theta	se	ci_95	p_value	reject_H0	note
(0, 1, 2, 3, 4, 12)	('In Progress_t0', 'Wait_t1', 'Closed_t2', 'Awaiting Assignment_t3', 'Assigned_t4', 'τ_t12')	949	0, 1, 2, 3, 4	0	In Progress_t0	Assigned_t4, Awaiting Assignment_t3, Closed_t2, Wait_t1	-1.7918	0.439988	[-2.6542, -0.9294]	0.0	True	θ=-1.7918: 1-unit ↑ util ↓ P(t0/In Progress_t0) by 179.2%
(0, 1, 2, 3, 4, 12)	('In Progress_t0', 'Wait_t1', 'Closed_t2', 'Awaiting Assignment_t3', 'Assigned_t4', 'τ_t12')	949	0, 1, 2, 3, 4	1	Wait_t1	Assigned_t4, Awaiting Assignment_t3, Closed_t2, In Progress_t0	-0.523356	0.480699	[-1.4655, 0.4188]	0.2763	False	θ=-0.5234: 1-unit ↑ util ↓ P(t1/Wait_t1) by 52.3%
(0, 1, 2, 3, 4, 12)	('In Progress_t0', 'Wait_t1', 'Closed_t2', 'Awaiting Assignment_t3', 'Assigned_t4', 'τ_t12')	949	0, 1, 2, 3, 4	2	Closed_t2	Assigned_t4, Awaiting Assignment_t3, In Progress_t0, Wait_t1	0.386316	0.506458	[-0.6063, 1.3790]	0.4456	False	θ=+0.3863: 1-unit ↑ util ↑ P(t2/Closed_t2) by 38.6%
(0, 1, 2, 3, 4, 12)	('In Progress_t0', 'Wait_t1', 'Closed_t2', 'Awaiting Assignment_t3', 'Assigned_t4', 'τ_t12')	949	0, 1, 2, 3, 4	3	Awaiting Assignment_t3	Assigned_t4, Closed_t2, In Progress_t0, Wait_t1	-0.536082	0.370486	[-1.2622, 0.1901]	0.1479	False	θ=-0.5361: 1-unit ↑ util ↓ P(t3/Awaiting Assignment_t3) by 53.6%
(0, 1, 2, 3, 4, 12)	('In Progress_t0', 'Wait_t1', 'Closed_t2', 'Awaiting Assignment_t3', 'Assigned_t4', 'τ_t12')	949	0, 1, 2, 3, 4	4	Assigned_t4	Awaiting Assignment_t3, Closed_t2, In Progress_t0, Wait_t1	-0.310652	0.346612	[-0.9900, 0.3687]	0.3701	False	θ=-0.3107: 1-unit ↑ util ↓ P(t4/Assigned_t4) by 31.1%
(5, 6, 7, 8, 9, 12)	('In Progress_t5', 'Wait_t6', 'Closed_t7', 'Awaiting Assignment_t8', 'Assigned_t9', 'τ_t12')	134	5, 6, 7, 8, 9	5	In Progress_t5	Assigned_t9, Awaiting Assignment_t8, Closed_t7, Wait_t6	-1.089647	0.899541	[-2.8527, 0.6735]	0.2258	False	θ=-1.0896: 1-unit ↑ util ↓ P(t5/In Progress_t5) by 109.0%
(5, 6, 7, 8, 9, 12)	('In Progress_t5', 'Wait_t6', 'Closed_t7', 'Awaiting Assignment_t8', 'Assigned_t9', 'τ_t12')	134	5, 6, 7, 8, 9	6	Wait_t6	Assigned_t9, Awaiting Assignment_t8, Closed_t7, In Progress_t5	-0.092014	0.288312	[-0.6571, 0.4731]	0.7496	False	θ=-0.0920: 1-unit ↑ util ↓ P(t6/Wait_t6) by 9.2%
(5, 6, 7, 8, 9, 12)	('In Progress_t5', 'Wait_t6', 'Closed_t7', 'Awaiting Assignment_t8', 'Assigned_t9', 'τ_t12')	134	5, 6, 7, 8, 9	7	Closed_t7	Assigned_t9, Awaiting Assignment_t8, In Progress_t5, Wait_t6	-0.115964	0.348083	[-0.7982, 0.5663]	0.739	False	θ=-0.1160: 1-unit ↑ util ↓ P(t7/Closed_t7) by 11.6%
(5, 6, 7, 8, 9, 12)	('In Progress_t5', 'Wait_t6', 'Closed_t7', 'Awaiting Assignment_t8', 'Assigned_t9', 'τ_t12')	134	5, 6, 7, 8, 9	8	Awaiting Assignment_t8	Assigned_t9, Closed_t7, In Progress_t5, Wait_t6	-0.957319	0.30393	[-1.5530, -0.3616]	0.0016	True	θ=-0.9573: 1-unit ↑ util ↓ P(t8/Awaiting Assignment_t8) by 95.7%
(5, 6, 7, 8, 9, 12)	('In Progress_t5', 'Wait_t6', 'Closed_t7', 'Awaiting Assignment_t8', 'Assigned_t9', 'τ_t12')	134	5, 6, 7, 8, 9	9	Assigned_t9	Awaiting Assignment_t8, Closed_t7, In Progress_t5, Wait_t6	-0.149017	0.11005	[-0.3647, 0.0667]	0.1757	False	θ=-0.1490: 1-unit ↑ util ↓ P(t9/Assigned_t9) by 14.9%
(10, 11, 12)	('In Progress_t10', 'Awaiting Assignment_t11', 'τ_t12')	273	10, 11	10	In Progress_t10	Awaiting Assignment_t11	-0.561449	0.286523	[-1.1230, 0.0001]	0.0501	False	θ=-0.5614: 1-unit ↑ util ↓ P(t10/In Progress_t10) by 56.1%
(10, 11, 12)	('In Progress_t10', 'Awaiting Assignment_t11', 'τ_t12')	273	10, 11	11	Awaiting Assignment_t11	In Progress_t10	-0.333504	0.183656	[-0.6935, 0.0265]	0.0694	False	θ=-0.3335: 1-unit ↑ util ↓ P(t11/Awaiting Assignment_t11) by 33.4%
(13, 14, 15, 16, 17)	('In Progress_t13', 'Wait_t14', 'Closed_t15', 'Awaiting Assignment_t16', 'Assigned_t17')	819	13, 14, 15, 16, 17	13	In Progress_t13	Assigned_t17, Awaiting Assignment_t16, Closed_t15, Wait_t14	-0.114529	0.220573	[-0.5469, 0.3178]	0.6036	False	θ=-0.1145: 1-unit ↑ util ↓ P(t13/In Progress_t13) by 11.5%
(13, 14, 15, 16, 17)	('In Progress_t13', 'Wait_t14', 'Closed_t15', 'Awaiting Assignment_t16', 'Assigned_t17')	819	13, 14, 15, 16, 17	14	Wait_t14	Assigned_t17, Awaiting Assignment_t16, Closed_t15, In Progress_t13	-0.044771	0.036709	[-0.1167, 0.0272]	0.2226	False	θ=-0.0448: 1-unit ↑ util ↓ P(t14/Wait_t14) by 4.5%
(13, 14, 15, 16, 17)	('In Progress_t13', 'Wait_t14', 'Closed_t15', 'Awaiting Assignment_t16', 'Assigned_t17')	819	13, 14, 15, 16, 17	15	Closed_t15	Assigned_t17, Awaiting Assignment_t16, In Progress_t13, Wait_t14	-0.013608	0.023462	[-0.0596, 0.0324]	0.5619	False	θ=-0.0136: 1-unit ↑ util ↓ P(t15/Closed_t15) by 1.4%
(13, 14, 15, 16, 17)	('In Progress_t13', 'Wait_t14', 'Closed_t15', 'Awaiting Assignment_t16', 'Assigned_t17')	819	13, 14, 15, 16, 17	16	Awaiting Assignment_t16	Assigned_t17, Closed_t15, In Progress_t13, Wait_t14	-0.07023	0.212117	[-0.4860, 0.3455]	0.7406	False	θ=-0.0702: 1-unit ↑ util ↓ P(t16/Awaiting Assignment_t16) by 7.0%
(13, 14, 15, 16, 17)	('In Progress_t13', 'Wait_t14', 'Closed_t15', 'Awaiting Assignment_t16', 'Assigned_t17')	819	13, 14, 15, 16, 17	17	Assigned_t17	Awaiting Assignment_t16, Closed_t15, In Progress_t13, Wait_t14	-0.019666	0.064339	[-0.1458, 0.1064]	0.7599	False	θ=-0.0197: 1-unit ↑ util ↓ P(t17/Assigned_t17) by 2.0%

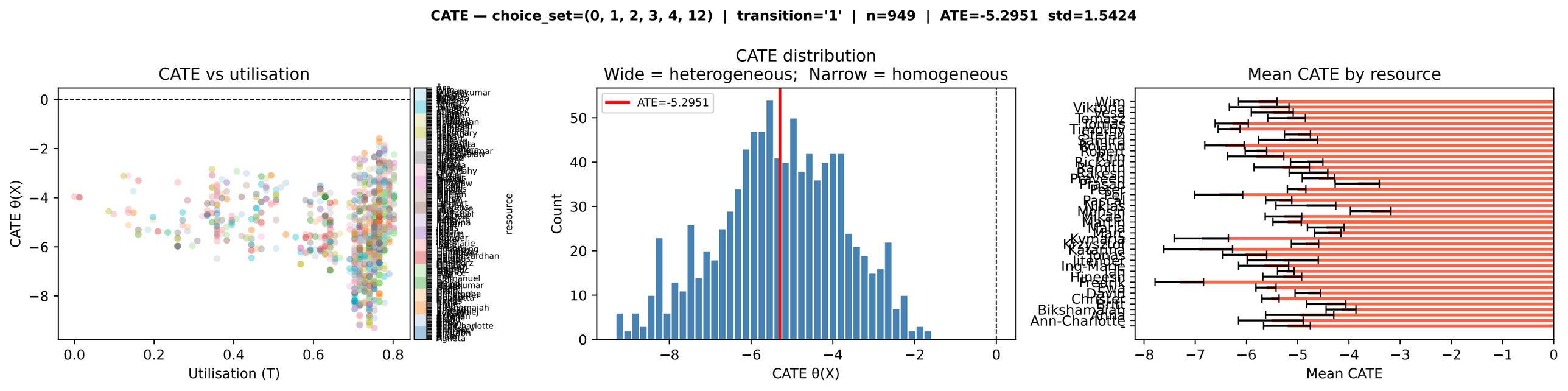
Step 3 — Causal Forest DML (ATE per transition)
ATE = mean of per-sample CATEs. Green rows = H0 rejected (p<0.05).
Less powerful for significance than Double ML, but basis for heterogeneity analysis.

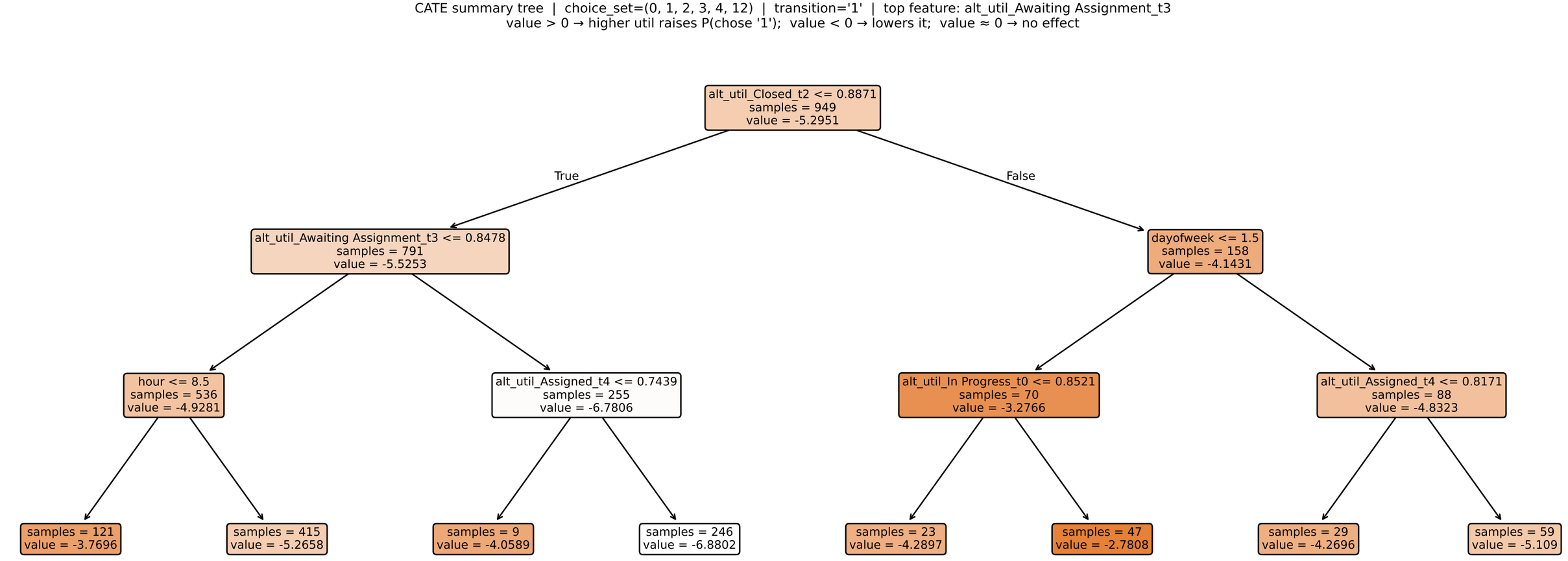
choice_set	choice_set_labels	n	choices_seen	transition	transition_label	alt_utils	ate	ci_95	p_value	reject_H0	
(0, 1, 2, 3, 4, 12)	('In Progress_t0', 'Wait_t1', 'Closed_t2', 'Awaiting Assignment_t3', 'Assigned_t4', 'τ_t12')	949	0, 1, 2, 3, 4	0	In Progress_t0	Assigned_t4, Awaiting Assignment_t3, Closed_t2, Wait_t1	-5.828992	[-10.1355, -1.5225]	0.008	True	ATE=-5.8290: 0→1 u n Progress_t0) by 582.9%
(0, 1, 2, 3, 4, 12)	('In Progress_t0', 'Wait_t1', 'Closed_t2', 'Awaiting Assignment_t3', 'Assigned_t4', 'τ_t12')	949	0, 1, 2, 3, 4	1	Wait_t1	Assigned_t4, Awaiting Assignment_t3, Closed_t2, In Progress_t0	-5.295144	[-10.3999, -0.1904]	0.042	True	ATE=-5.2951: 0→ t1/Wait_t1) by 529.5%
(0, 1, 2, 3, 4, 12)	('In Progress_t0', 'Wait_t1', 'Closed_t2', 'Awaiting Assignment_t3', 'Assigned_t4', 'τ_t12')	949	0, 1, 2, 3, 4	2	Closed_t2	Assigned_t4, Awaiting Assignment_t3, In Progress_t0, Wait_t1	-4.256705	[-10.0168, 1.5034]	0.1475	False	ATE=-4.2567: 0→1 /Closed_t2) by 425.7%
(0, 1, 2, 3, 4, 12)	('In Progress_t0', 'Wait_t1', 'Closed_t2', 'Awaiting Assignment_t3', 'Assigned_t4', 'τ_t12')	949	0, 1, 2, 3, 4	3	Awaiting Assignment_t3	Assigned_t4, Closed_t2, In Progress_t0, Wait_t1	-2.457923	[-5.8734, 0.9575]	0.1584	False	ATE=-2.4579: 0→1 util ↓ ng Assignment_t3) by 245.8%
(0, 1, 2, 3, 4, 12)	('In Progress_t0', 'Wait_t1', 'Closed_t2', 'Awaiting Assignment_t3', 'Assigned_t4', 'τ_t12')	949	0, 1, 2, 3, 4	4	Assigned_t4	Awaiting Assignment_t3, Closed_t2, In Progress_t0, Wait_t1	-0.476254	[-3.9588, 3.0063]	0.7887	False	ATE=-0.4763: 0→1 /Assigned_t4) by 47.6%
(5, 6, 7, 8, 9, 12)	('In Progress_t5', 'Wait_t6', 'Closed_t7', 'Awaiting Assignment_t8', 'Assigned_t9', 'τ_t12')	134	5, 6, 7, 8, 9	5	In Progress_t5	Assigned_t9, Awaiting Assignment_t8, Closed_t7, Wait_t6	-2.088639	[-4.5789, 0.4016]	0.1002	False	ATE=-2.0886: 0→1 u n Progress_t5) by 208.9%
(5, 6, 7, 8, 9, 12)	('In Progress_t5', 'Wait_t6', 'Closed_t7', 'Awaiting Assignment_t8', 'Assigned_t9', 'τ_t12')	134	5, 6, 7, 8, 9	6	Wait_t6	Assigned_t9, Awaiting Assignment_t8, Closed_t7, In Progress_t5	0.423645	[-0.3211, 1.1684]	0.2649	False	ATE=+0.4236: 0 (t6/Wait_t6) by 42.4%
(5, 6, 7, 8, 9, 12)	('In Progress_t5', 'Wait_t6', 'Closed_t7', 'Awaiting Assignment_t8', 'Assigned_t9', 'τ_t12')	134	5, 6, 7, 8, 9	7	Closed_t7	Assigned_t9, Awaiting Assignment_t8, In Progress_t5, Wait_t6	-1.960342	[-4.2025, 0.2818]	0.0866	False	ATE=-1.9603: 0→1 /Closed_t7) by 196.0%
(5, 6, 7, 8, 9, 12)	('In Progress_t5', 'Wait_t6', 'Closed_t7', 'Awaiting Assignment_t8', 'Assigned_t9', 'τ_t12')	134	5, 6, 7, 8, 9	8	Awaiting Assignment_t8	Assigned_t9, Closed_t7, In Progress_t5, Wait_t6	-1.046883	[-2.0526, -0.0412]	0.0413	True	ATE=-1.0469: 0→1 util ↓ ng Assignment_t8) by 104.7%
(5, 6, 7, 8, 9, 12)	('In Progress_t5', 'Wait_t6', 'Closed_t7', 'Awaiting Assignment_t8', 'Assigned_t9', 'τ_t12')	134	5, 6, 7, 8, 9	9	Assigned_t9	Awaiting Assignment_t8, Closed_t7, In Progress_t5, Wait_t6	-0.263324	[-0.9876, 0.4609]	0.4761	False	ATE=-0.2633: 0→1 /Assigned_t9) by 26.3%
(10, 11, 12)	('In Progress_t10', 'Awaiting Assignment_t11', 'τ_t12')	273	10, 11	10	In Progress_t10	Awaiting Assignment_t11	-0.415249	[-1.2293, 0.3988]	0.3174	False	ATE=-0.4152: 0→1 u n Progress_t10) by 41.5%
(10, 11, 12)	('In Progress_t10', 'Awaiting Assignment_t11', 'τ_t12')	273	10, 11	11	Awaiting Assignment_t11	In Progress_t10	-0.234641	[-1.1438, 0.6746]	0.613	False	ATE=-0.2346: 0→1 util ↓ ing Assignment_t11) by 23.5%
(13, 14, 15, 16, 17)	('In Progress_t13', 'Wait_t14', 'Closed_t15', 'Awaiting Assignment_t16', 'Assigned_t17')	819	13, 14, 15, 16, 17	13	In Progress_t13	Assigned_t17, Awaiting Assignment_t16, Closed_t15, Wait_t14	-0.133493	[-2.8045, 2.5375]	0.922	False	ATE=-0.1335: 0→1 u n Progress_t13) by 13.3%
(13, 14, 15, 16, 17)	('In Progress_t13', 'Wait_t14', 'Closed_t15', 'Awaiting Assignment_t16', 'Assigned_t17')	819	13, 14, 15, 16, 17	14	Wait_t14	Assigned_t17, Awaiting Assignment_t16, Closed_t15, In Progress_t13	-0.308369	[-1.0501, 0.4334]	0.4152	False	ATE=-0.3084: 0→ 14/Wait_t14) by 30.8%
(13, 14, 15, 16, 17)	('In Progress_t13', 'Wait_t14', 'Closed_t15', 'Awaiting Assignment_t16', 'Assigned_t17')	819	13, 14, 15, 16, 17	15	Closed_t15	Assigned_t17, Awaiting Assignment_t16, In Progress_t13, Wait_t14	0.000131	[-0.0036, 0.0038]	0.9444	False	ATE=+0.0001: 0→ 15/Closed_t15) by 0.0%
(13, 14, 15, 16, 17)	('In Progress_t13', 'Wait_t14', 'Closed_t15', 'Awaiting Assignment_t16', 'Assigned_t17')	819	13, 14, 15, 16, 17	16	Awaiting Assignment_t16	Assigned_t17, Closed_t15, In Progress_t13, Wait_t14	-0.189537	[-2.5930, 2.2139]	0.8772	False	ATE=-0.1895: 0→1 util ↓ ing Assignment_t16) by 19.0%
(13, 14, 15, 16, 17)	('In Progress_t13', 'Wait_t14', 'Closed_t15', 'Awaiting Assignment_t16', 'Assigned_t17')	819	13, 14, 15, 16, 17	17	Assigned_t17	Awaiting Assignment_t16, Closed_t15, In Progress_t13, Wait_t14	-0.300239	[-1.2087, 0.6082]	0.5171	False	ATE=-0.3002: 0→1 u /Assigned_t17) by 30.0%

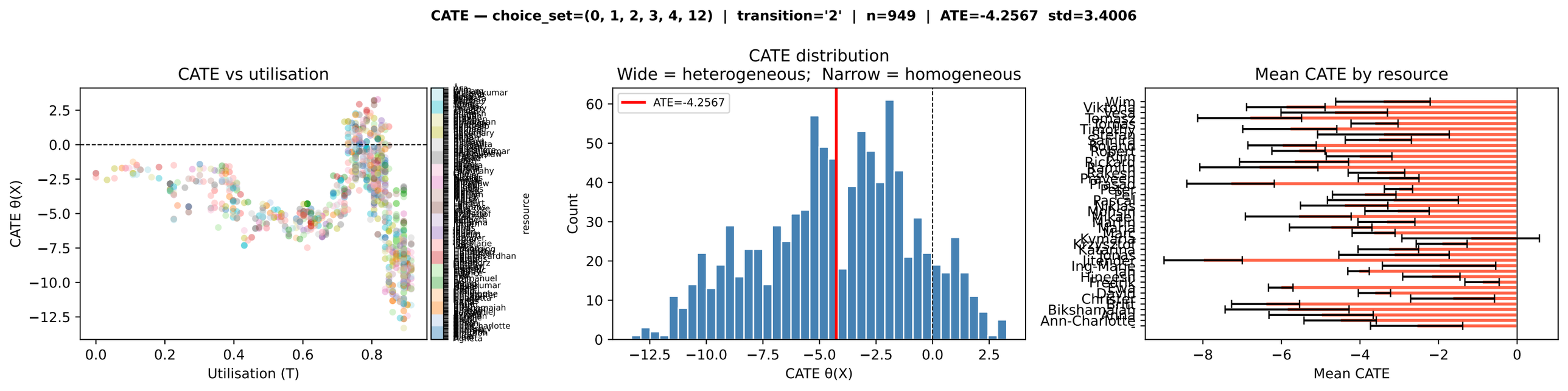


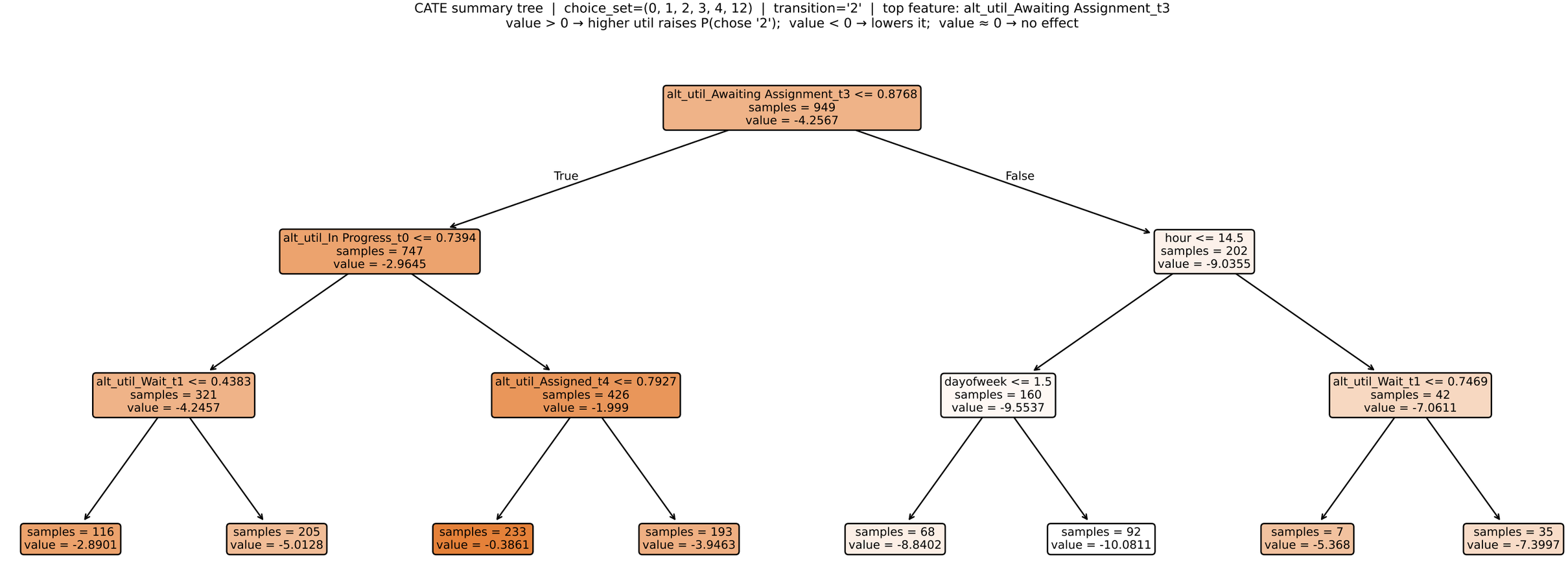

CATE — choice_set=(0, 1, 2, 3, 4, 12) | transition='0' | n=949 | ATE=-5.8290 std=2.0437
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=-5.8290
Count
CATE θ(X)
Mean CATE by resource
Mean CATE

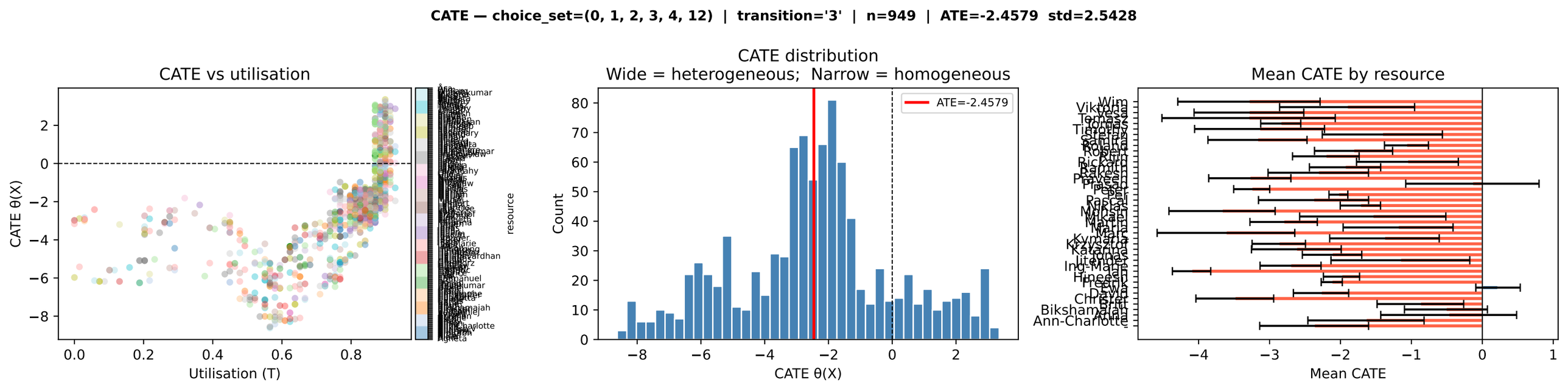

CATE summary tree | choice_set=(0, 1, 2, 3, 4, 12) | transition='0' | top feature: alt_util_Closed_t2
value > 0 → higher util raises P(chose '0'); value < 0 → lowers it; value ≈ 0 → no effect
alt_util_Closed_t2 <= 0.7366
samples = 949
value = -5.829
True
False
alt_util_Closed_t2 <= 0.6613
samples = 337
value = -3.7963
alt_util_Awaiting Assignment_t3 <= 0.8913
samples = 612
value = -6.9483
alt_util_Awaiting Assignment_t3 <= 0.4565
samples = 264
value = -3.4452
hour <= 11.5
samples = 73
value = -5.0663
hour <= 12.5
samples = 497
value = -6.5004
alt_util_Wait_t1 <= 0.7716
samples = 115
value = -8.8837
samples = 86
value = -4.7495
samples = 178
value = -2.815
samples = 34
value = -5.5467
samples = 39
value = -4.6475
samples = 276
value = -7.0094
samples = 221
value = -5.8648
samples = 32
value = -7.5454
samples = 83
value = -9.3997

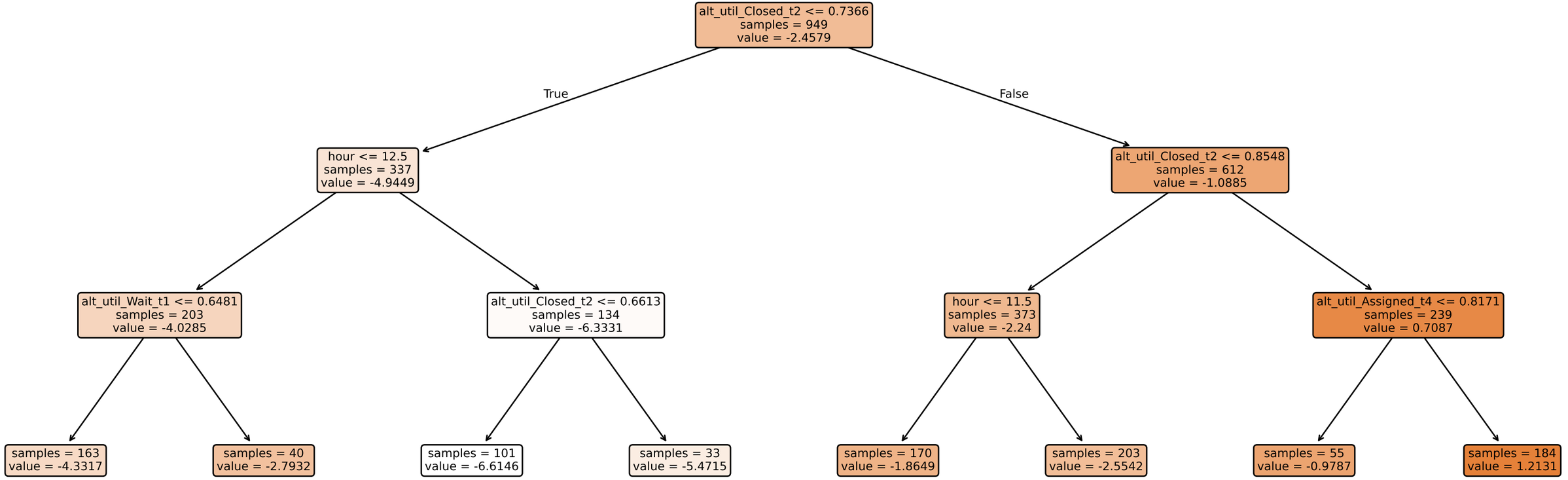

CATE — choice_set=(0, 1, 2, 3, 4, 12) | transition='1' | n=949 | ATE=-5.2951 std=1.5424
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=-5.2951
Count
CATE θ(X)
Mean CATE by resource
Mean CATE


CATE summary tree | choice_set=(0, 1, 2, 3, 4, 12) | transition='1' | top feature: alt_util_Awaiting Assignment_t3
value > 0 → higher util raises P(chose '1'); value < 0 → lowers it; value ≈ 0 → no effect
alt_util_Closed_t2 <= 0.8871
samples = 949
value = -5.2951
True
False
alt_util_Awaiting Assignment_t3 <= 0.8478
samples = 791
value = -5.5253
dayofweek <= 1.5
samples = 158
value = -4.1431
hour <= 8.5
samples = 536
value = -4.9281
alt_util_Assigned_t4 <= 0.7439
samples = 255
value = -6.7806
alt_util_In Progress_t0 <= 0.8521
samples = 70
value = -3.2766
alt_util_Assigned_t4 <= 0.8171
samples = 88
value = -4.8323
samples = 121
value = -3.7696
samples = 415
value = -5.2658
samples = 9
value = -4.0589
samples = 246
value = -6.8802
samples = 23
value = -4.2897
samples = 47
value = -2.7808
samples = 29
value = -4.2696
samples = 59
value = -5.109

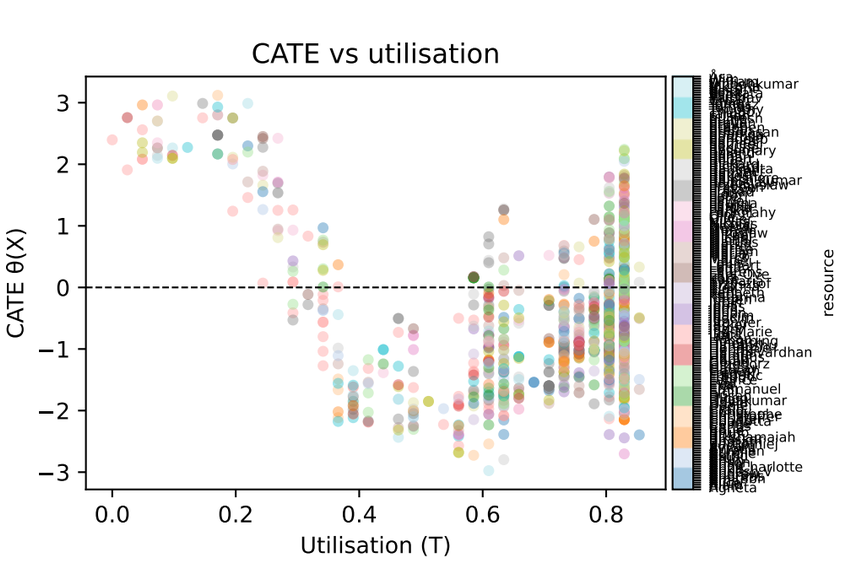

CATE — choice_set=(0, 1, 2, 3, 4, 12) | transition='2' | n=949 | ATE=-4.2567 std=3.4006
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=-4.2567
Count
CATE θ(X)
Mean CATE by resource
Mean CATE

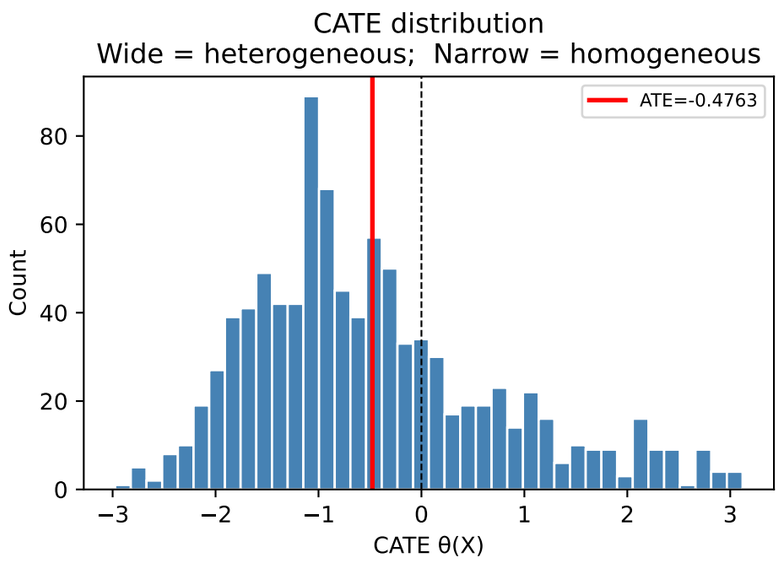

CATE summary tree | choice_set=(0, 1, 2, 3, 4, 12) | transition='2' | top feature: alt_util_Awaiting Assignment_t3
value > 0 → higher util raises P(chose '2'); value < 0 → lowers it; value ≈ 0 → no effect
alt_util_Awaiting Assignment_t3 <= 0.8768
samples = 949
value = -4.2567
True
False
alt_util_In Progress_t0 <= 0.7394
samples = 747
value = -2.9645
hour <= 14.5
samples = 202
value = -9.0355
alt_util_Wait_t1 <= 0.4383
samples = 321
value = -4.2457
alt_util_Assigned_t4 <= 0.7927
samples = 426
value = -1.999
dayofweek <= 1.5
samples = 160
value = -9.5537
alt_util_Wait_t1 <= 0.7469
samples = 42
value = -7.0611
samples = 116
value = -2.8901
samples = 205
value = -5.0128
samples = 233
value = -0.3861
samples = 193
value = -3.9463
samples = 68
value = -8.8402
samples = 92
value = -10.0811
samples = 7
value = -5.368
samples = 35
value = -7.3997

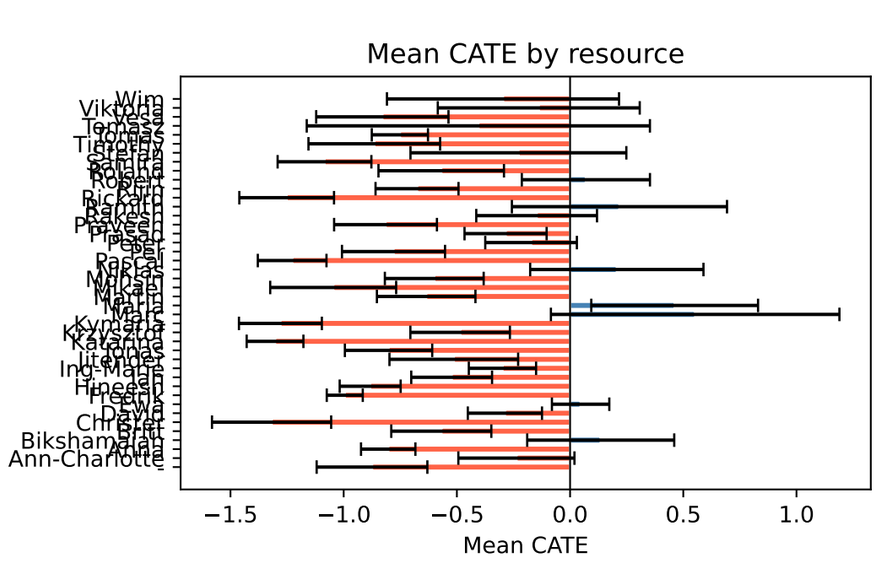

CATE — choice_set=(0, 1, 2, 3, 4, 12) | transition='3' | n=949 | ATE=-2.4579 std=2.5428
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=-2.4579
Count
CATE θ(X)
Mean CATE by resource
Mean CATE

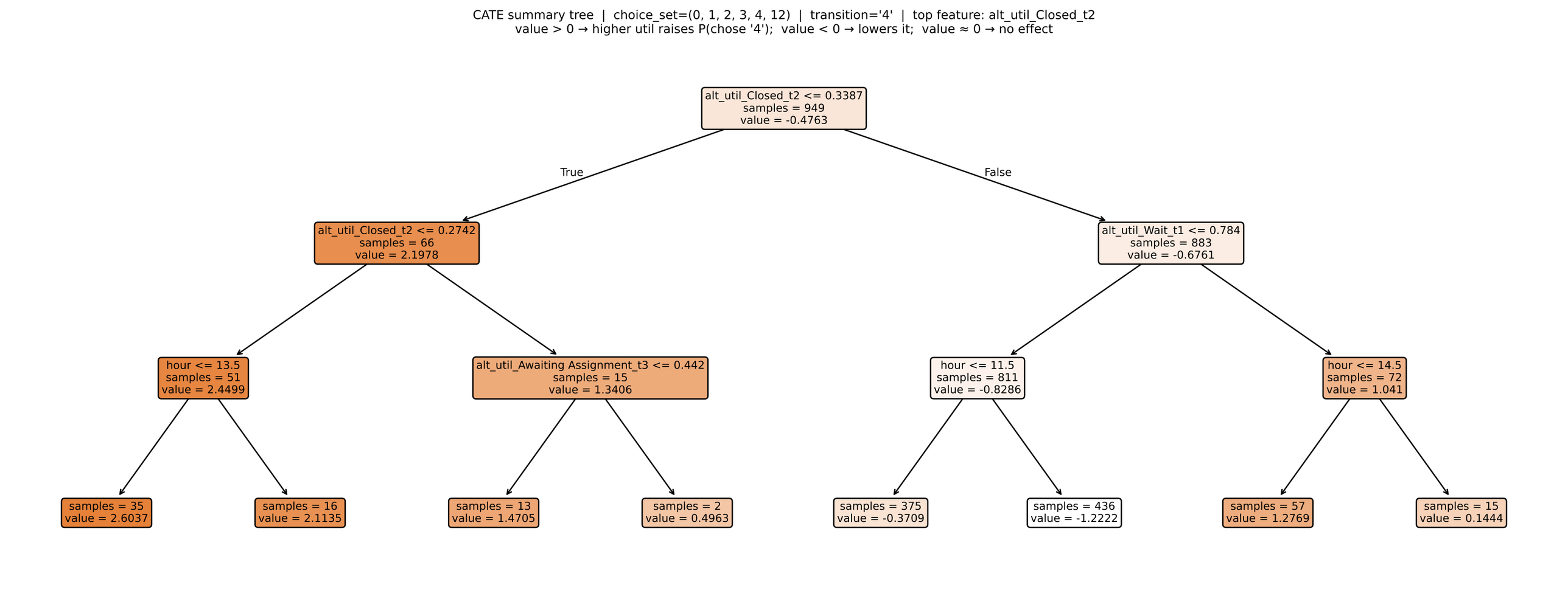

CATE summary tree | choice_set=(0, 1, 2, 3, 4, 12) | transition='3' | top feature: alt_util_Closed_t2
value > 0 → higher util raises P(chose '3'); value < 0 → lowers it; value ≈ 0 → no effect
alt_util_Closed_t2 <= 0.7366
samples = 949
value = -2.4579
True
False
hour <= 12.5
samples = 337
value = -4.9449
alt_util_Closed_t2 <= 0.8548
samples = 612
value = -1.0885
alt_util_Wait_t1 <= 0.6481
samples = 203
value = -4.0285
alt_util_Closed_t2 <= 0.6613
samples = 134
value = -6.3331
hour <= 11.5
samples = 373
value = -2.24
alt_util_Assigned_t4 <= 0.8171
samples = 239
value = 0.7087
samples = 163
value = -4.3317
samples = 40
value = -2.7932
samples = 101
value = -6.6146
samples = 33
value = -5.4715
samples = 170
value = -1.8649
samples = 203
value = -2.5542
samples = 55
value = -0.9787
samples = 184
value = 1.2131

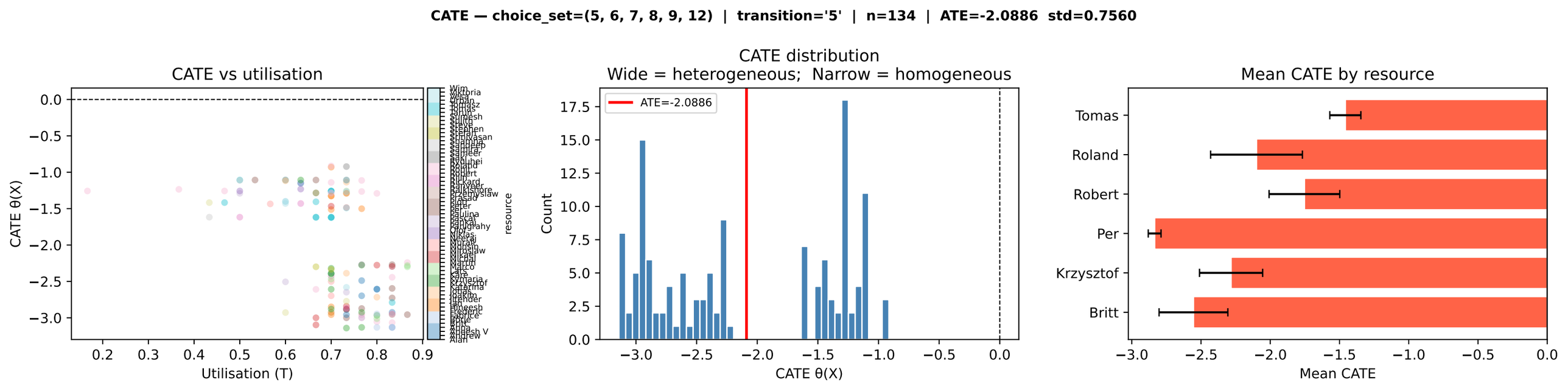
CATE — choice_set=(0, 1, 2, 3, 4, 12) | transition='4' | n=949 | ATE=-0.4763 std=1.1984

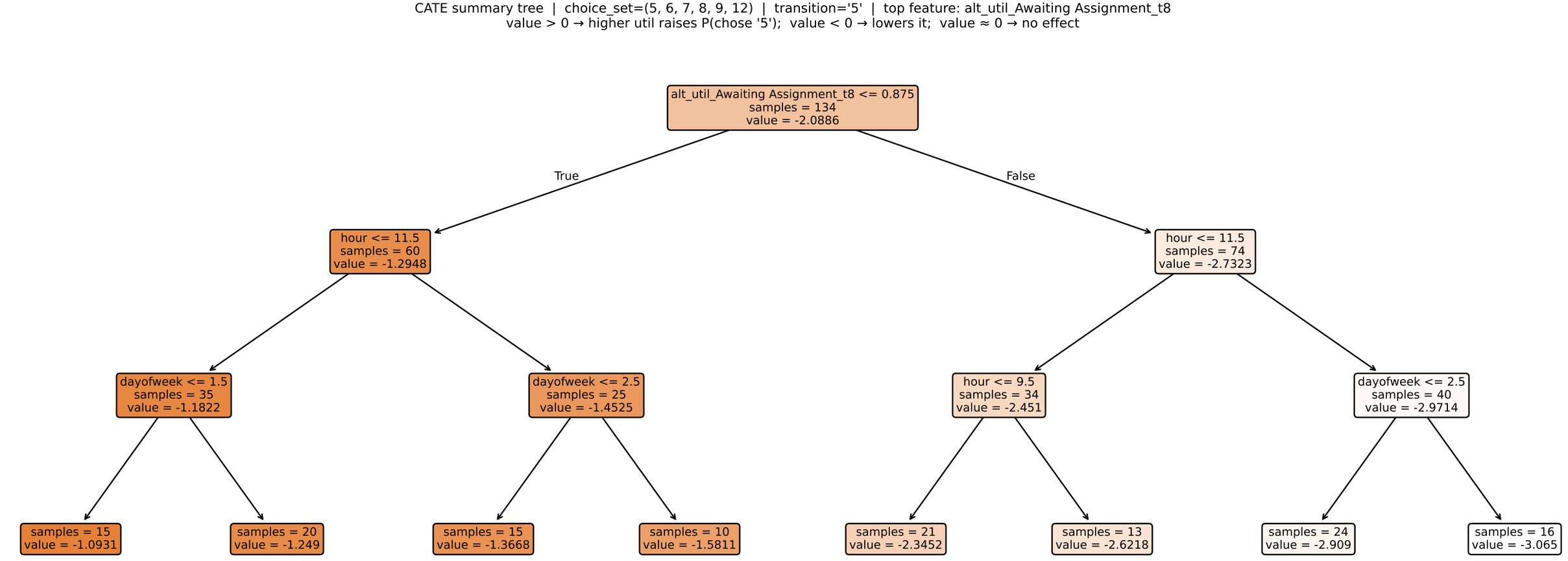


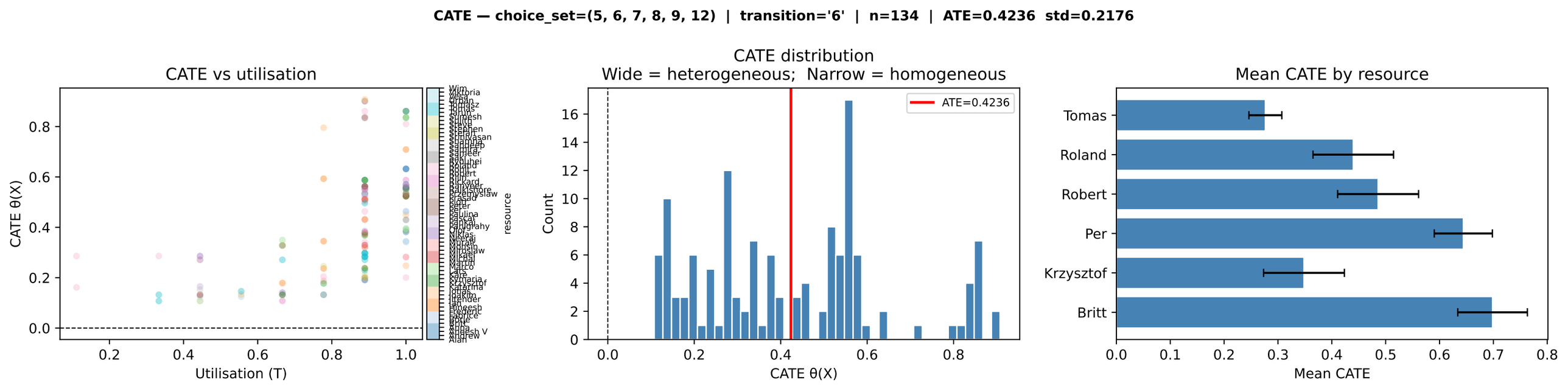


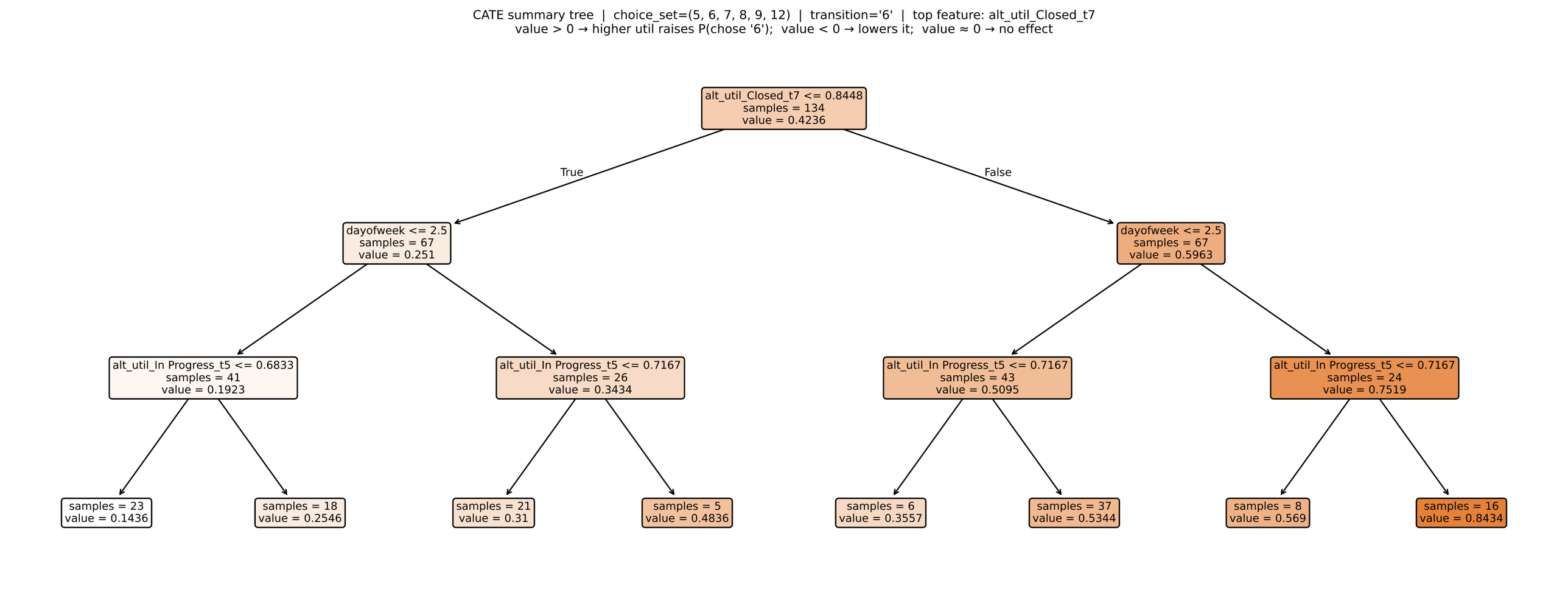


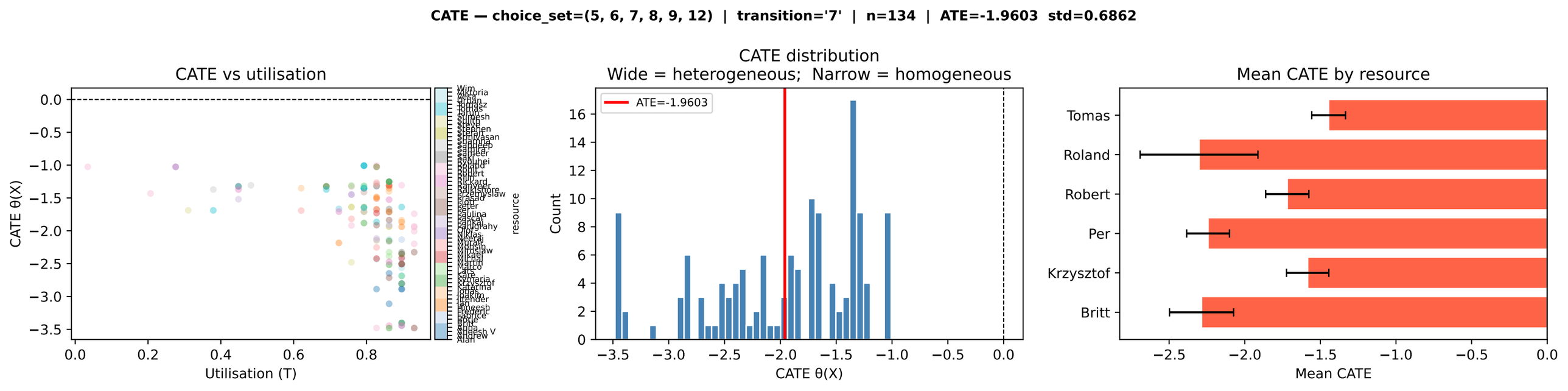

CATE summary tree | choice_set=(0, 1, 2, 3, 4, 12) | transition='4' | top feature: alt_util_Closed_t2
value > 0 → higher util raises P(chose '4'); value < 0 → lowers it; value ≈ 0 → no effect
alt_util_Closed_t2 <= 0.3387
samples = 949
value = -0.4763
True
False
alt_util_Closed_t2 <= 0.2742
samples = 66
value = 2.1978
alt_util_Wait_t1 <= 0.784
samples = 883
value = -0.6761
hour <= 13.5
samples = 51
value = 2.4499
alt_util_Awaiting Assignment_t3 <= 0.442
samples = 15
value = 1.3406
hour <= 11.5
samples = 811
value = -0.8286
hour <= 14.5
samples = 72
value = 1.041
samples = 35
value = 2.6037
samples = 16
value = 2.1135
samples = 13
value = 1.4705
samples = 2
value = 0.4963
samples = 375
value = -0.3709
samples = 436
value = -1.2222
samples = 57
value = 1.2769
samples = 15
value = 0.1444

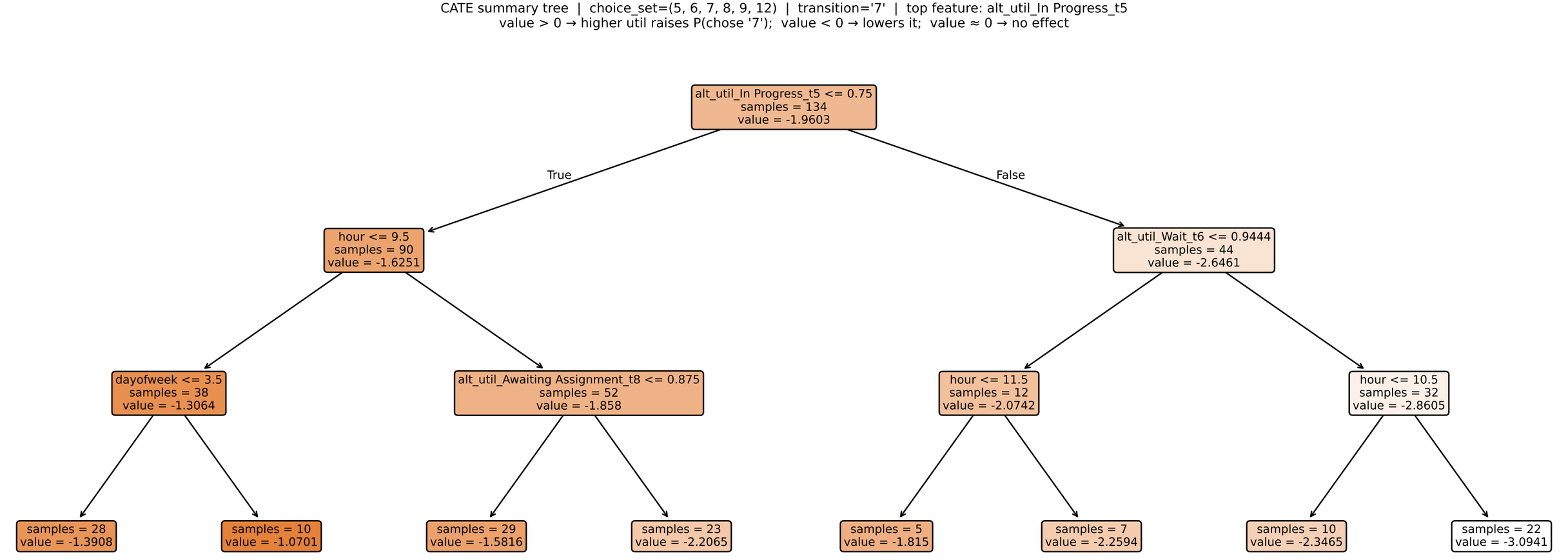

CATE — choice_set=(5, 6, 7, 8, 9, 12) | transition='5' | n=134 | ATE=-2.0886 std=0.7560
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=-2.0886
Count
CATE θ(X)
Mean CATE by resource
Tomas
Roland
Robert
Per
Krzysztof
Britt
Mean CATE

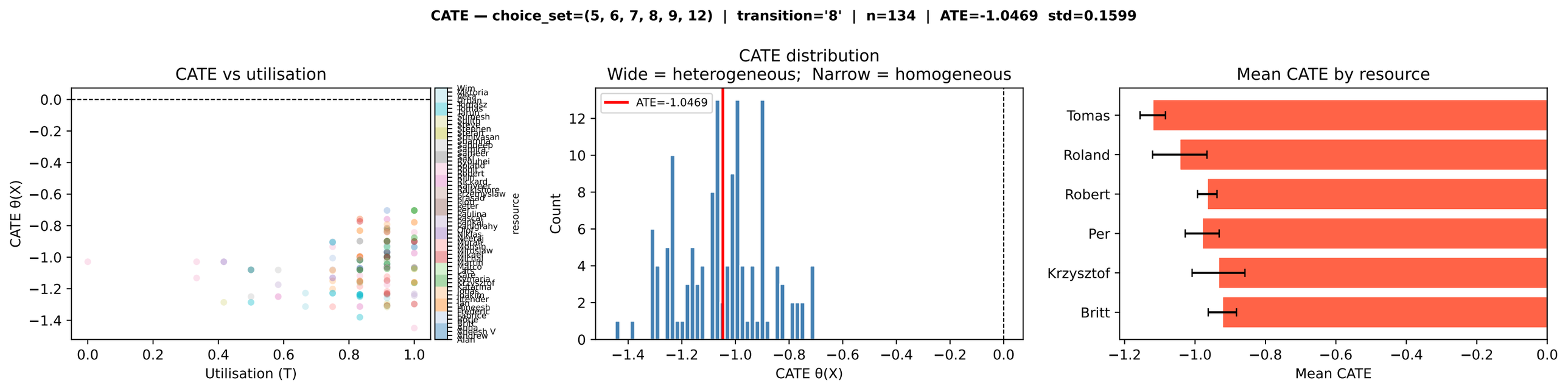

CATE summary tree | choice_set=(5, 6, 7, 8, 9, 12) | transition='5' | top feature: alt_util_Awaiting Assignment_t8
value > 0 → higher util raises P(chose '5'); value < 0 → lowers it; value ≈ 0 → no effect
alt_util_Awaiting Assignment_t8 <= 0.875
samples = 134
value = -2.0886
True
False
hour <= 11.5
samples = 60
value = -1.2948
hour <= 11.5
samples = 74
value = -2.7323
dayofweek <= 1.5
samples = 35
value = -1.1822
dayofweek <= 2.5
samples = 25
value = -1.4525
hour <= 9.5
samples = 34
value = -2.451
dayofweek <= 2.5
samples = 40
value = -2.9714
samples = 15
value = -1.0931
samples = 20
value = -1.249
samples = 15
value = -1.3668
samples = 10
value = -1.5811
samples = 21
value = -2.3452
samples = 13
value = -2.6218
samples = 24
value = -2.909
samples = 16
value = -3.065

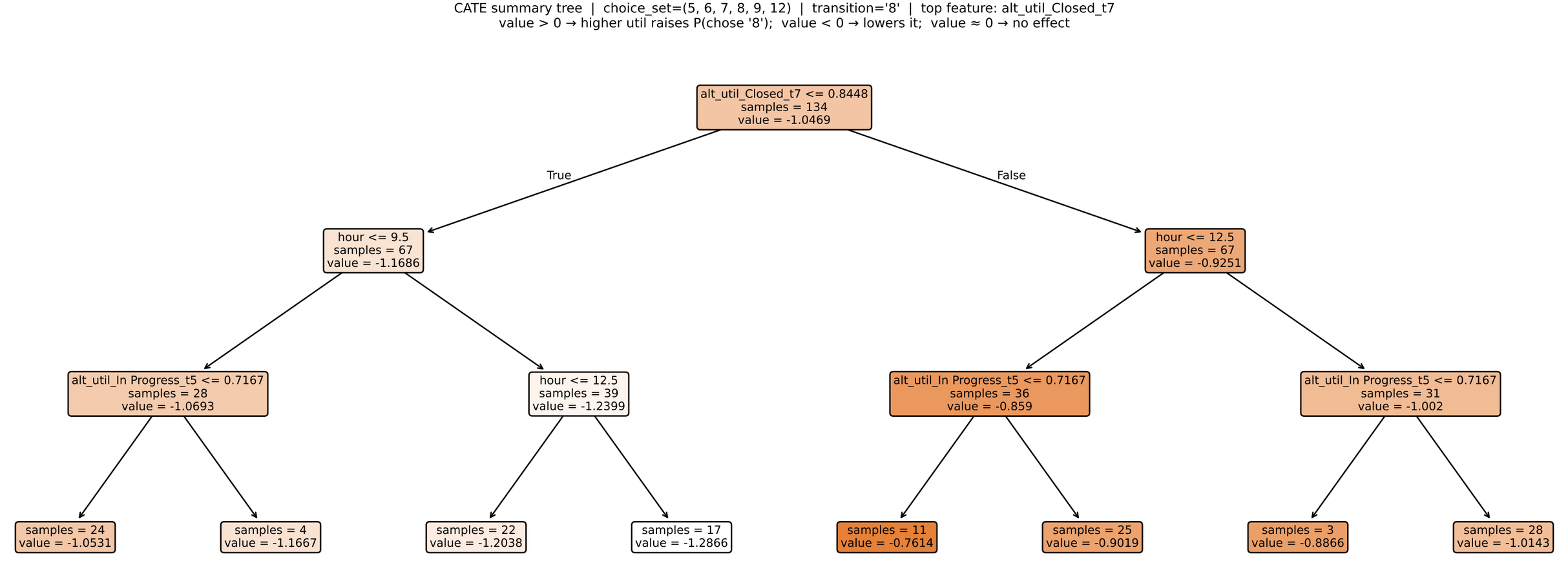

CATE — choice_set=(5, 6, 7, 8, 9, 12) | transition='6' | n=134 | ATE=0.4236 std=0.2176
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=0.4236
Count
CATE θ(X)
Mean CATE by resource
Tomas
Roland
Robert
Per
Krzysztof
Britt
Mean CATE

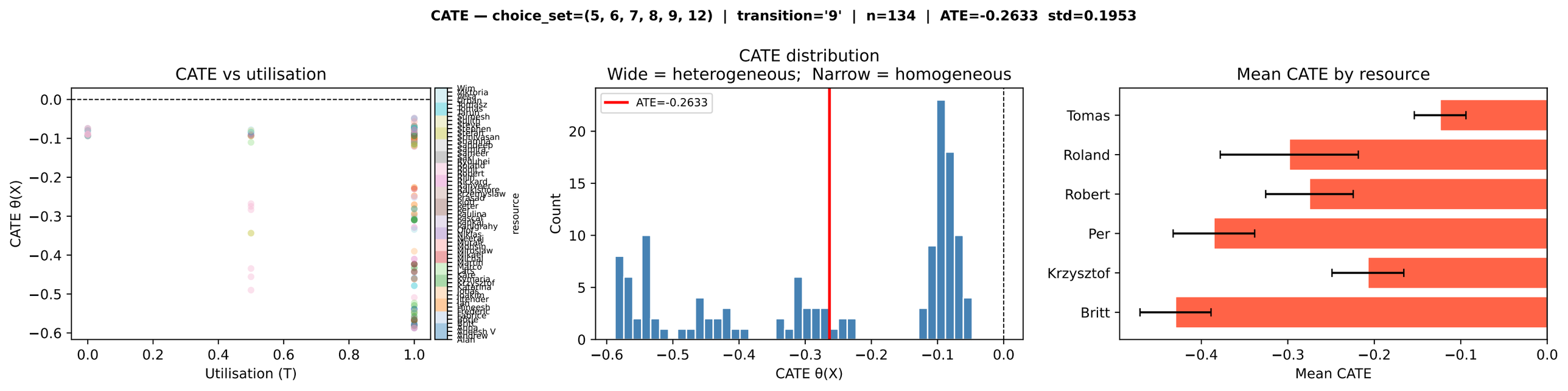

CATE summary tree | choice_set=(5, 6, 7, 8, 9, 12) | transition='6' | top feature: alt_util_Closed_t7
value > 0 → higher util raises P(chose '6'); value < 0 → lowers it; value ≈ 0 → no effect
alt_util_Closed_t7 <= 0.8448
samples = 134
value = 0.4236
True
False
dayofweek <= 2.5
samples = 67
value = 0.251
dayofweek <= 2.5
samples = 67
value = 0.5963
alt_util_ln Progress_t5 <= 0.6833
samples = 41
value = 0.1923
alt_util_ln Progress_t5 <= 0.7167
samples = 26
value = 0.3434
alt_util_ln Progress_t5 <= 0.7167
samples = 43
value = 0.5095
alt_util_ln Progress_t5 <= 0.7167
samples = 24
value = 0.7519
samples = 23
value = 0.1436
samples = 18
value = 0.2546
samples = 21
value = 0.31
samples = 5
value = 0.4836
samples = 6
value = 0.3557
samples = 37
value = 0.5344
samples = 8
value = 0.569
samples = 16
value = 0.8434

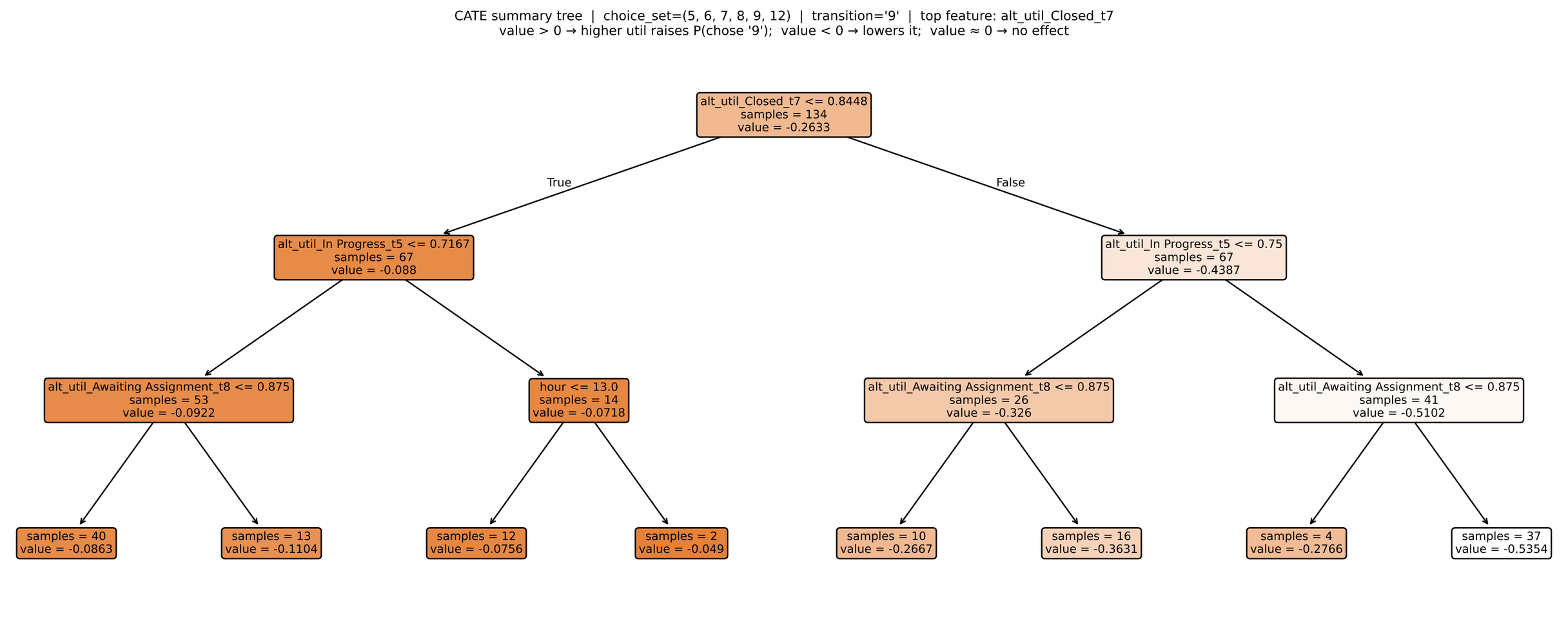

CATE — choice_set=(5, 6, 7, 8, 9, 12) | transition='7' | n=134 | ATE=-1.9603 std=0.6862
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=-1.9603
Count
CATE θ(X)
Mean CATE by resource
Tomas
Roland
Robert
Per
Krzysztof
Britt
Mean CATE

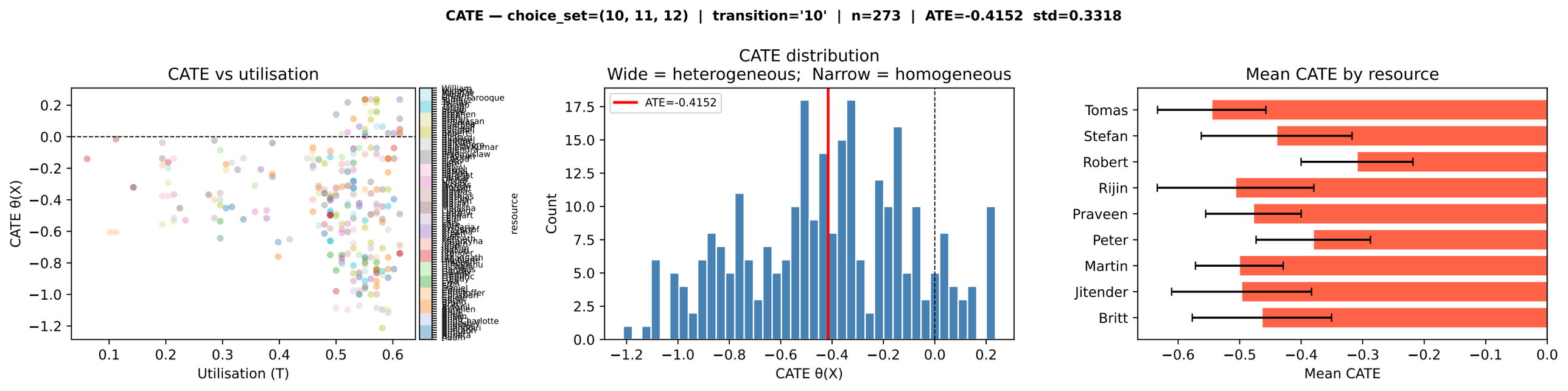

CATE summary tree | choice_set=(5, 6, 7, 8, 9, 12) | transition='7' | top feature: alt_util_In Progress_t5
value > 0 → higher util raises P(chose '7'); value < 0 → lowers it; value ≈ 0 → no effect
alt_util_In Progress_t5 <= 0.75
samples = 134
value = -1.9603
True
False
hour <= 9.5
samples = 90
value = -1.6251
alt_util_Wait_t6 <= 0.9444
samples = 44
value = -2.6461
dayofweek <= 3.5
samples = 38
value = -1.3064
alt_util_Awaiting Assignment_t8 <= 0.875
samples = 52
value = -1.858
hour <= 11.5
samples = 12
value = -2.0742
hour <= 10.5
samples = 32
value = -2.8605
samples = 28
value = -1.3908
samples = 10
value = -1.0701
samples = 29
value = -1.5816
samples = 23
value = -2.2065
samples = 5
value = -1.815
samples = 7
value = -2.2594
samples = 10
value = -2.3465
samples = 22
value = -3.0941

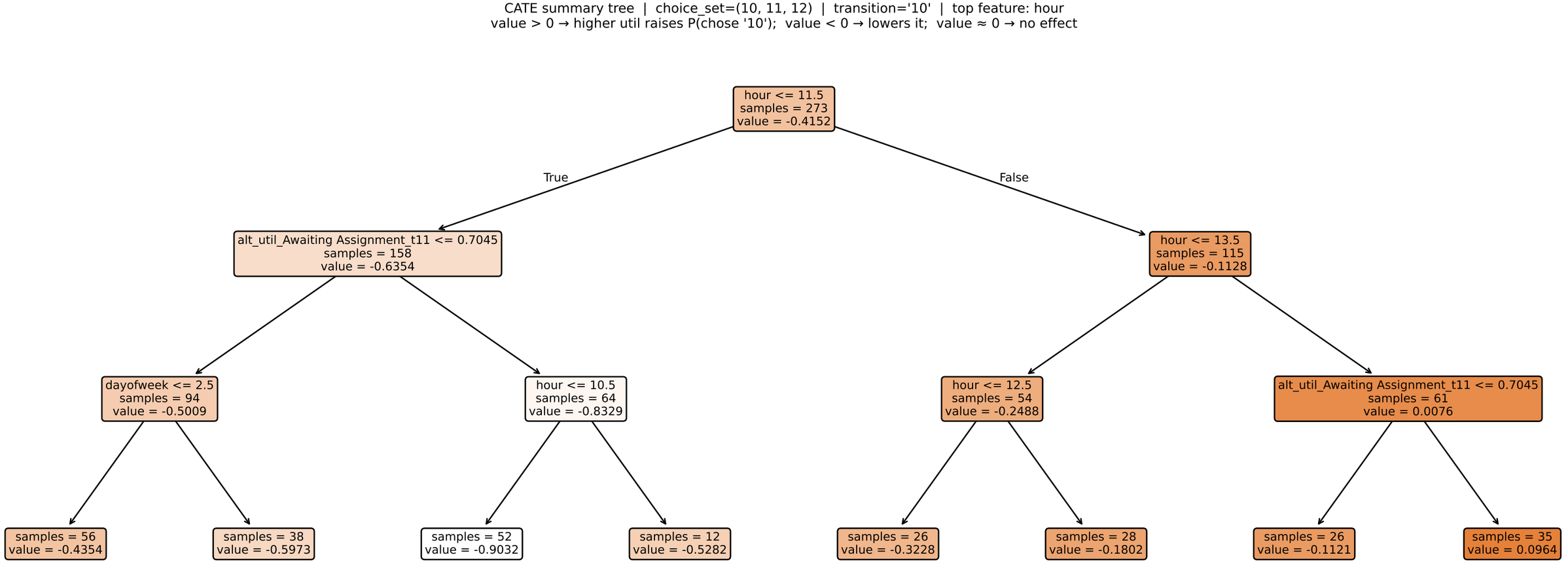

CATE — choice_set=(5, 6, 7, 8, 9, 12) | transition='8' | n=134 | ATE=-1.0469 std=0.1599
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=-1.0469
Count
CATE θ(X)
Mean CATE by resource
Tomas
Roland
Robert
Per
Krzysztof
Britt
Mean CATE

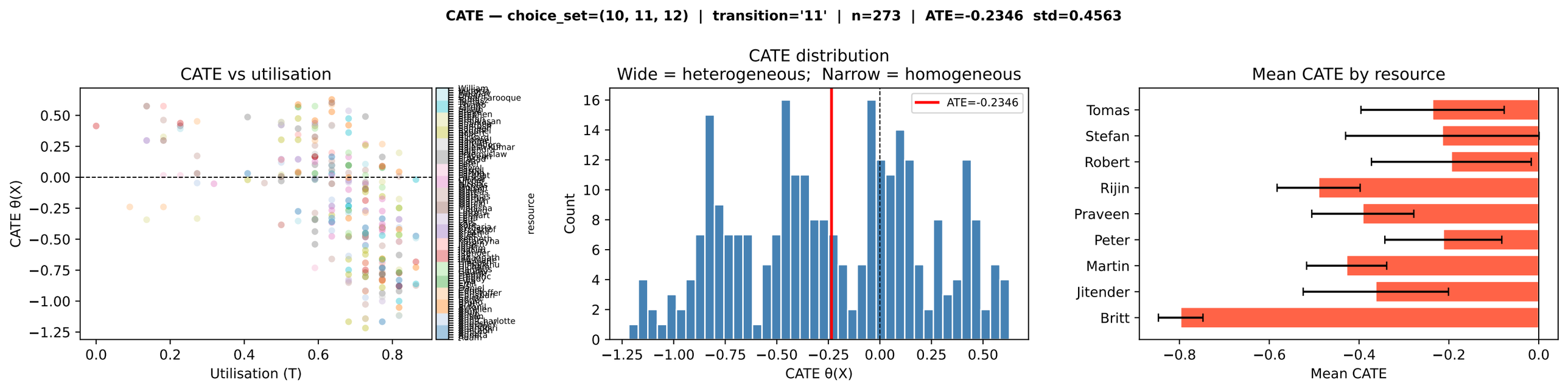

CATE summary tree | choice_set=(5, 6, 7, 8, 9, 12) | transition='8' | top feature: alt_util_Closed_t7
value > 0 → higher util raises P(chose '8'); value < 0 → lowers it; value ≈ 0 → no effect
alt_util_Closed_t7 <= 0.8448
samples = 134
value = -1.0469
True
False
hour <= 9.5
samples = 67
value = -1.1686
hour <= 12.5
samples = 67
value = -0.9251
alt_util_In Progress_t5 <= 0.7167
samples = 28
value = -1.0693
hour <= 12.5
samples = 39
value = -1.2399
alt_util_In Progress_t5 <= 0.7167
samples = 36
value = -0.859
alt_util_In Progress_t5 <= 0.7167
samples = 31
value = -1.002
samples = 24
value = -1.0531
samples = 4
value = -1.1667
samples = 22
value = -1.2038
samples = 17
value = -1.2866
samples = 11
value = -0.7614
samples = 25
value = -0.9019
samples = 3
value = -0.8866
samples = 28
value = -1.0143

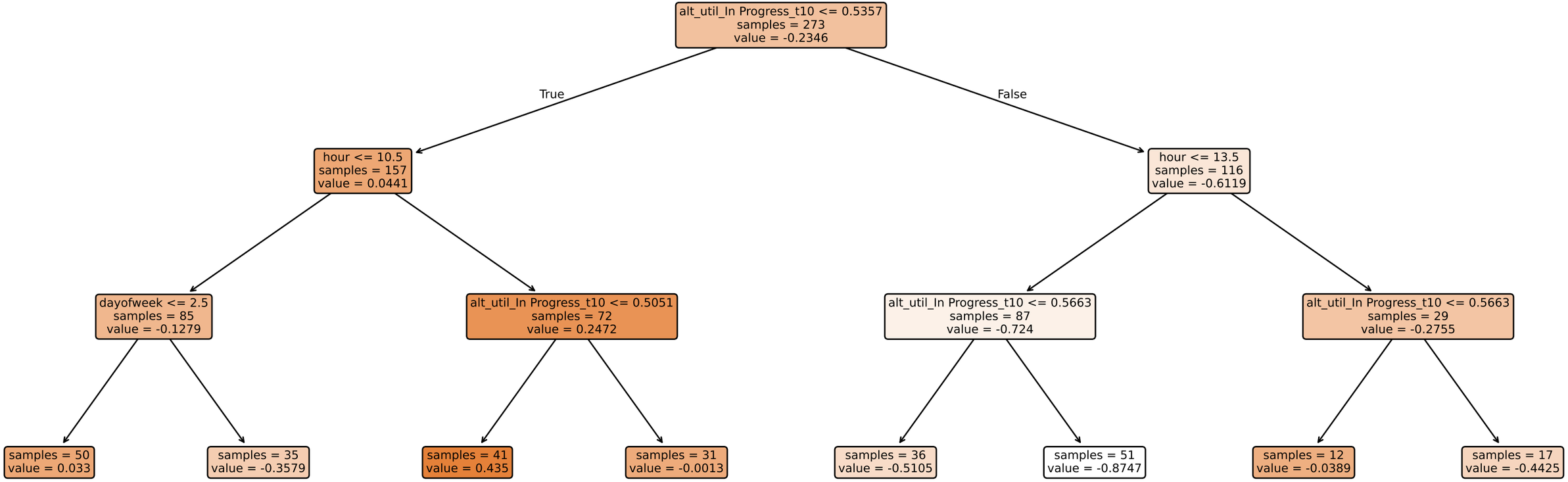

CATE — choice_set=(5, 6, 7, 8, 9, 12) | transition='9' | n=134 | ATE=-0.2633 std=0.1953
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=-0.2633
Count
CATE θ(X)
Mean CATE by resource
Tomas
Roland
Robert
Per
Krzysztof
Britt
Mean CATE

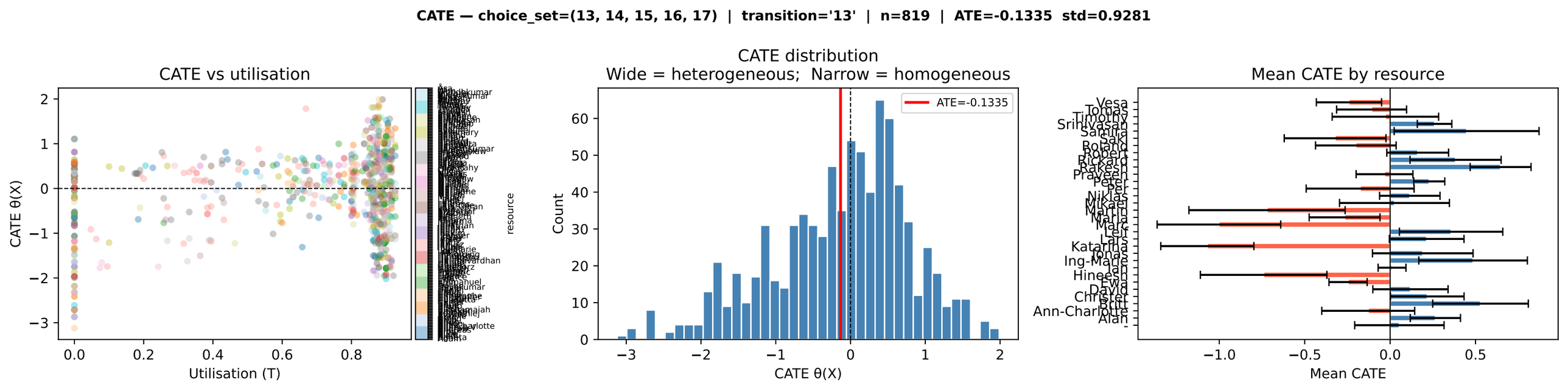

CATE summary tree | choice_set=(5, 6, 7, 8, 9, 12) | transition='9' | top feature: alt_util_Closed_t7
value > 0 → higher util raises P(chose '9'); value < 0 → lowers it; value ≈ 0 → no effect
alt_util_Closed_t7 <= 0.8448
samples = 134
value = -0.2633
True
False
alt_util_In Progress_t5 <= 0.7167
samples = 67
value = -0.088
alt_util_In Progress_t5 <= 0.75
samples = 67
value = -0.4387
alt_util_Awaiting Assignment_t8 <= 0.875
samples = 53
value = -0.0922
hour <= 13.0
samples = 14
value = -0.0718
alt_util_Awaiting Assignment_t8 <= 0.875
samples = 26
value = -0.326
alt_util_Awaiting Assignment_t8 <= 0.875
samples = 41
value = -0.5102
samples = 40
value = -0.0863
samples = 13
value = -0.1104
samples = 12
value = -0.0756
samples = 2
value = -0.049
samples = 10
value = -0.2667
samples = 16
value = -0.3631
samples = 4
value = -0.2766
samples = 37
value = -0.5354

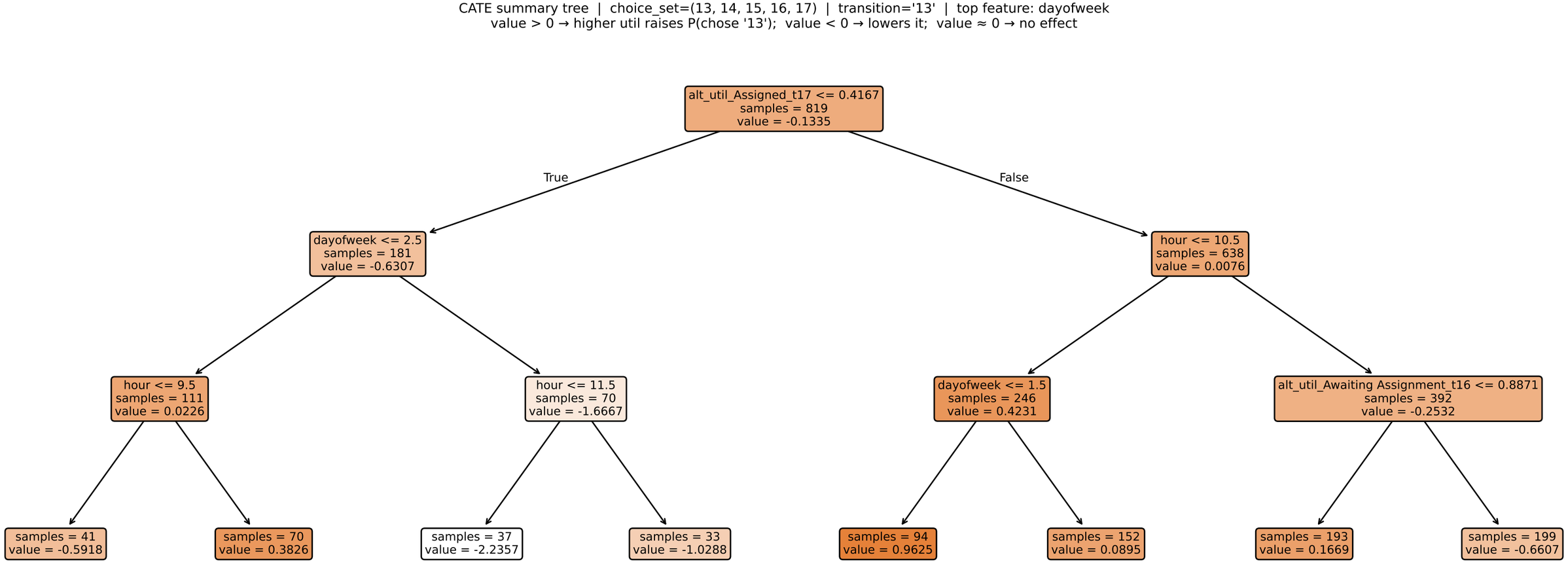

CATE — choice_set=(10, 11, 12) | transition='10' | n=273 | ATE=-0.4152 std=0.3318
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=-0.4152
Count
CATE θ(X)
Mean CATE by resource
Tomas
Stefan
Robert
Rijin
Praveen
Peter
Martin
Jitender
Britt
Mean CATE

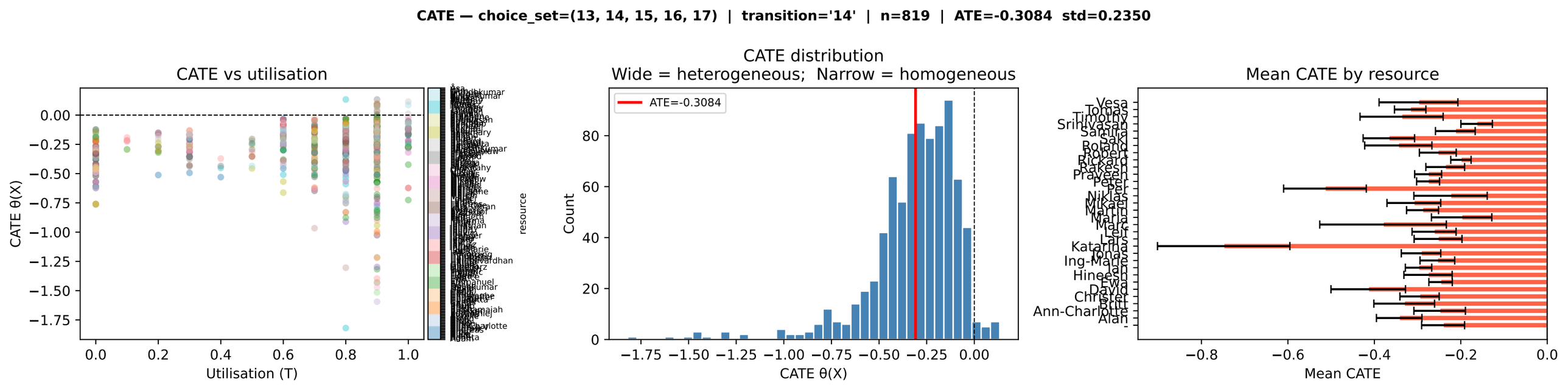

CATE summary tree | choice_set=(10, 11, 12) | transition='10' | top feature: hour
value > 0 → higher util raises P(chose '10'); value < 0 → lowers it; value ≈ 0 → no effect
hour <= 11.5
samples = 273
value = -0.4152
True
False
alt_util_Awaiting Assignment_t11 <= 0.7045
samples = 158
value = -0.6354
hour <= 13.5
samples = 115
value = -0.1128
dayofweek <= 2.5
samples = 94
value = -0.5009
hour <= 10.5
samples = 64
value = -0.8329
hour <= 12.5
samples = 54
value = -0.2488
alt_util_Awaiting Assignment_t11 <= 0.7045
samples = 61
value = 0.0076
samples = 56
value = -0.4354
samples = 38
value = -0.5973
samples = 52
value = -0.9032
samples = 12
value = -0.5282
samples = 26
value = -0.3228
samples = 28
value = -0.1802
samples = 26
value = -0.1121
samples = 35
value = 0.0964

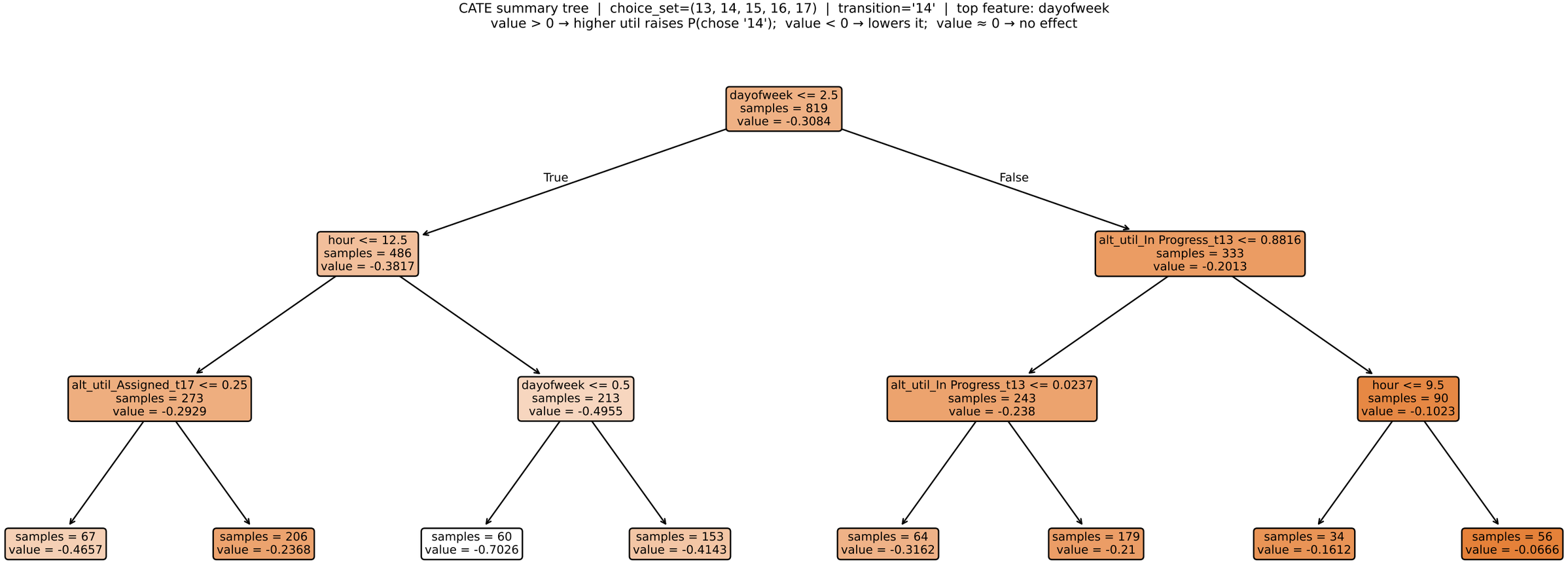

CATE — choice_set=(10, 11, 12) | transition='11' | n=273 | ATE=-0.2346 std=0.4563
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
Count
CATE θ(X)
ATE=-0.2346
Mean CATE by resource
Tomas
Stefan
Robert
Rijin
Praveen
Peter
Martin
Jitender
Britt
Mean CATE

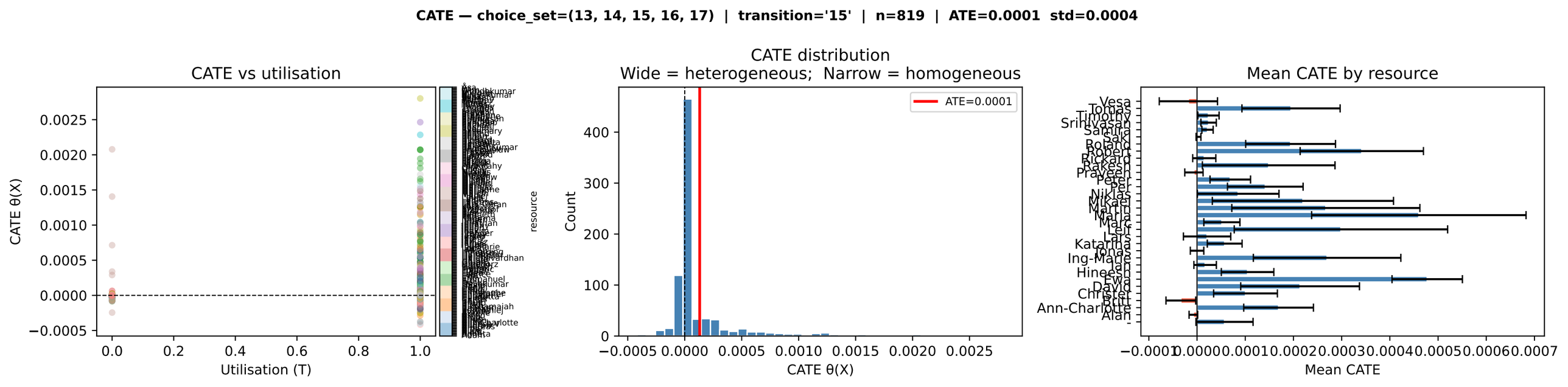

CATE summary tree | choice_set=(10, 11, 12) | transition='11' | top feature: alt_util_ln Progress_t10
value > 0 → higher util raises P(chose '11'); value < 0 → lowers it; value ≈ 0 → no effect
alt_util_ln Progress_t10 <= 0.5357
samples = 273
value = -0.2346
True
False
hour <= 10.5
samples = 157
value = 0.0441
hour <= 13.5
samples = 116
value = -0.6119
dayofweek <= 2.5
samples = 85
value = -0.1279
alt_util_ln Progress_t10 <= 0.5051
samples = 72
value = 0.2472
alt_util_ln Progress_t10 <= 0.5663
samples = 87
value = -0.724
alt_util_ln Progress_t10 <= 0.5663
samples = 29
value = -0.2755
samples = 50
value = 0.033
samples = 35
value = -0.3579
samples = 41
value = 0.435
samples = 31
value = -0.0013
samples = 36
value = -0.5105
samples = 51
value = -0.8747
samples = 12
value = -0.0389
samples = 17
value = -0.4425

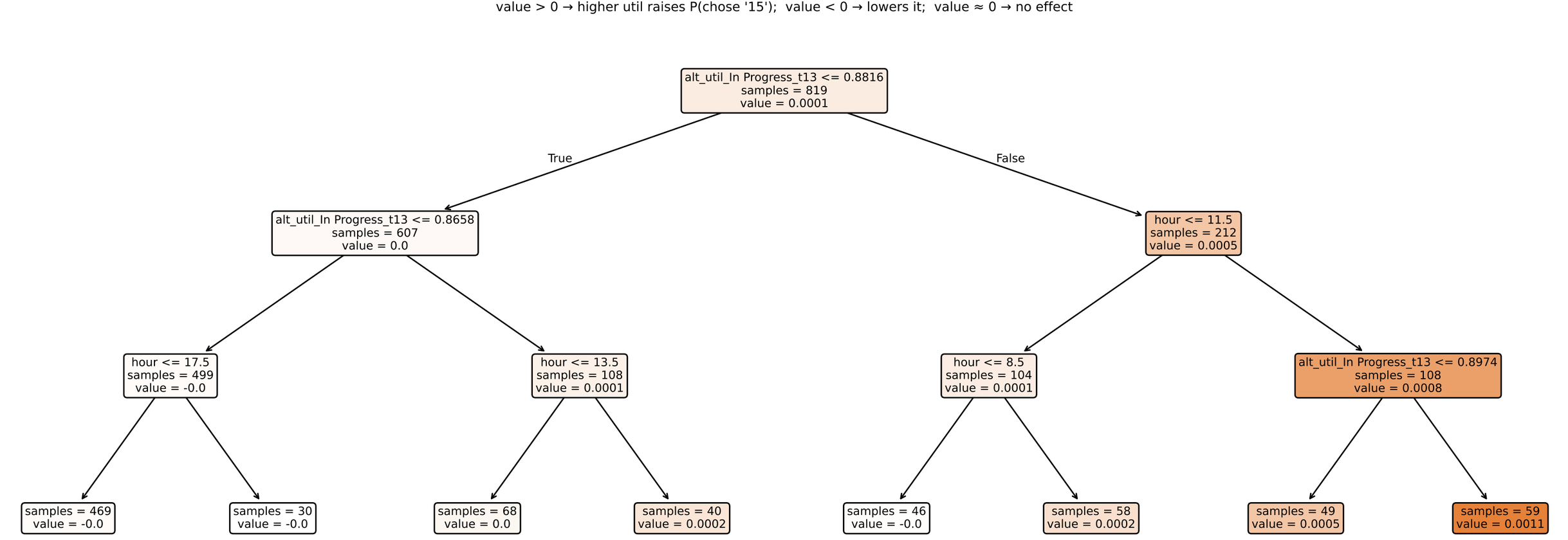

CATE — choice_set=(13, 14, 15, 16, 17) | transition='13' | n=819 | ATE=-0.1335 std=0.9281
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=-0.1335
Count
CATE θ(X)
Mean CATE by resource
Mean CATE

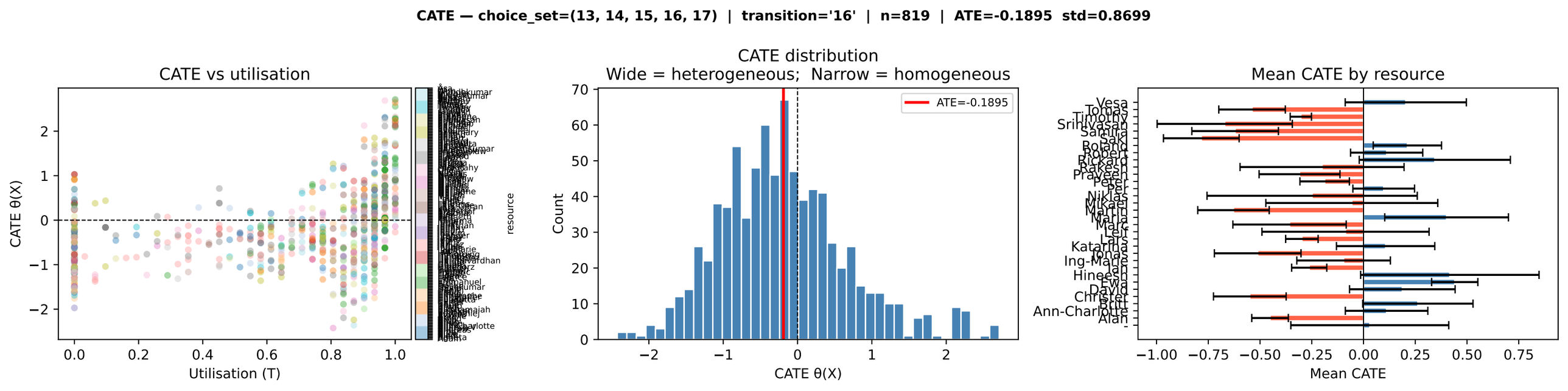

CATE summary tree | choice_set=(13, 14, 15, 16, 17) | transition='13' | top feature: dayofweek
value > 0 → higher util raises P(chose '13'); value < 0 → lowers it; value ≈ 0 → no effect
alt_util_Assigned_t17 <= 0.4167
samples = 819
value = -0.1335
True
False
dayofweek <= 2.5
samples = 181
value = -0.6307
hour <= 10.5
samples = 638
value = 0.0076
hour <= 9.5
samples = 111
value = 0.0226
hour <= 11.5
samples = 70
value = -1.6667
dayofweek <= 1.5
samples = 246
value = 0.4231
alt_util_Awaiting Assignment_t16 <= 0.8871
samples = 392
value = -0.2532
samples = 41
value = -0.5918
samples = 70
value = 0.3826
samples = 37
value = -2.2357
samples = 33
value = -1.0288
samples = 94
value = 0.9625
samples = 152
value = 0.0895
samples = 193
value = 0.1669
samples = 199
value = -0.6607

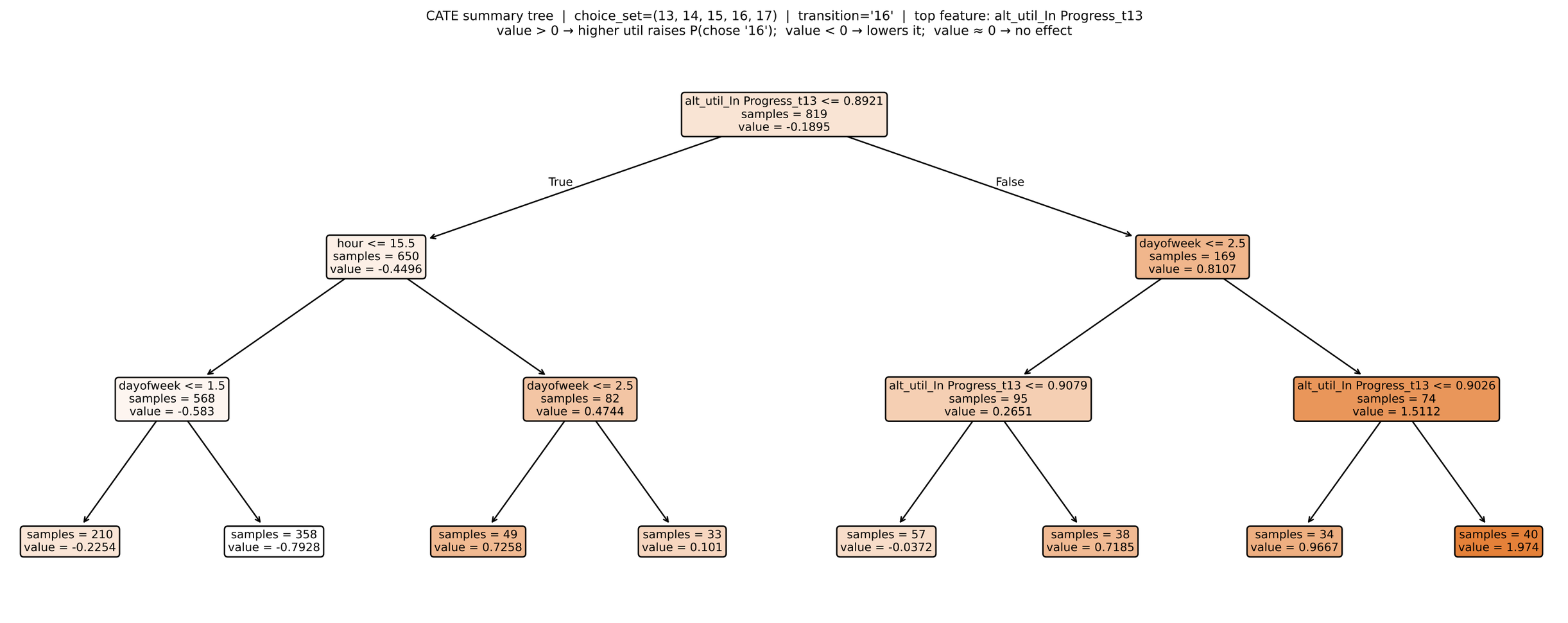

CATE — choice_set=(13, 14, 15, 16, 17) | transition='14' | n=819 | ATE=-0.3084 std=0.2350
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=-0.3084
Count
CATE θ(X)
Mean CATE by resource
Mean CATE

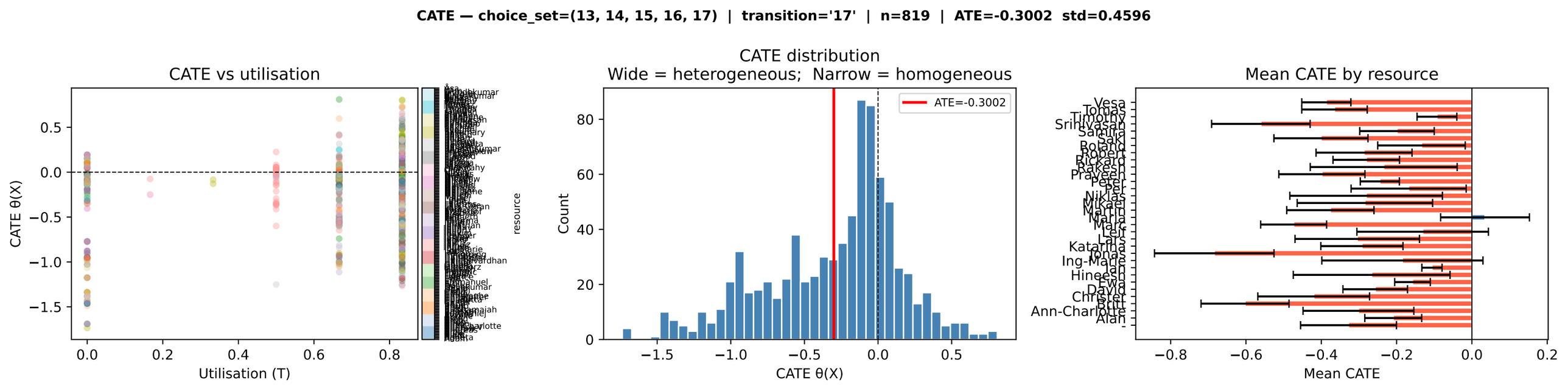

CATE summary tree | choice_set=(13, 14, 15, 16, 17) | transition='14' | top feature: dayofweek
value > 0 → higher util raises P(chose '14'); value < 0 → lowers it; value ≈ 0 → no effect
dayofweek <= 2.5
samples = 819
value = -0.3084
True
False
hour <= 12.5
samples = 486
value = -0.3817
alt_util_ln Progress_t13 <= 0.8816
samples = 333
value = -0.2013
alt_util_Assigned_t17 <= 0.25
samples = 273
value = -0.2929
dayofweek <= 0.5
samples = 213
value = -0.4955
alt_util_ln Progress_t13 <= 0.0237
samples = 243
value = -0.238
hour <= 9.5
samples = 90
value = -0.1023
samples = 67
value = -0.4657
samples = 206
value = -0.2368
samples = 60
value = -0.7026
samples = 153
value = -0.4143
samples = 64
value = -0.3162
samples = 179
value = -0.21
samples = 34
value = -0.1612
samples = 56
value = -0.0666

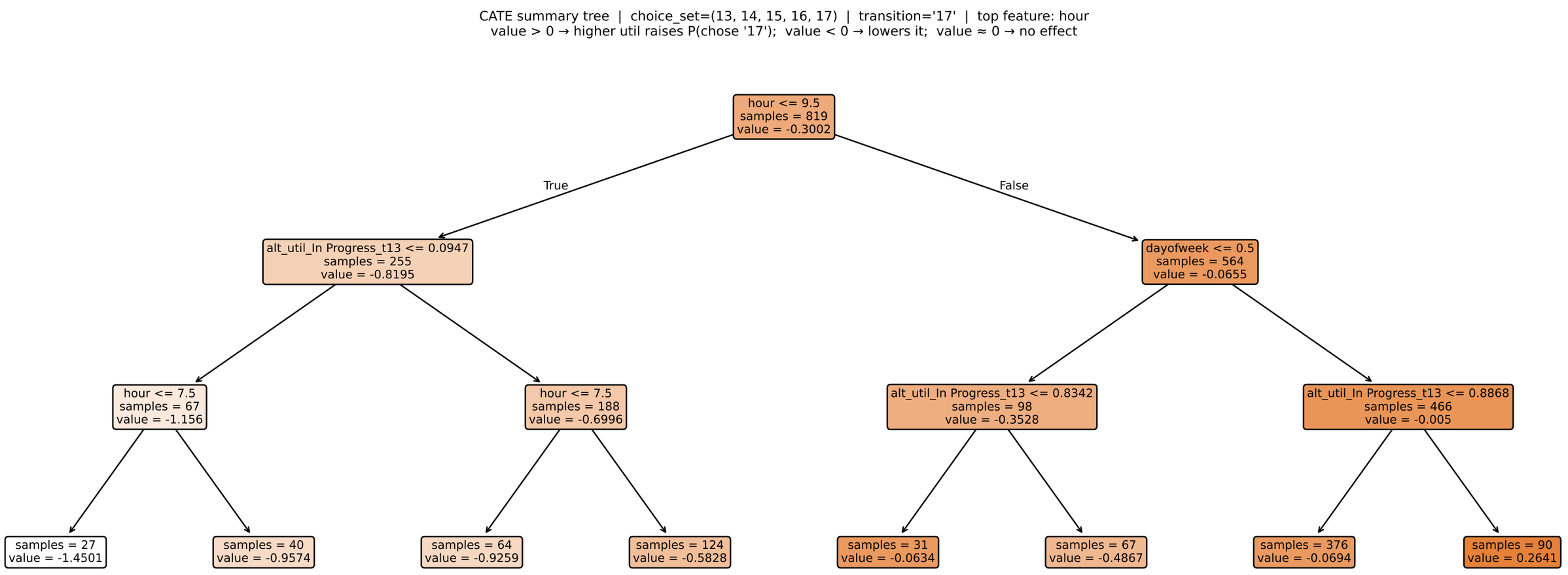

CATE — choice_set=(13, 14, 15, 16, 17) | transition='15' | n=819 | ATE=0.0001 std=0.0004
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=0.0001
Count
CATE θ(X)
Mean CATE by resource
Mean CATE


CATE summary tree | choice_set=(13, 14, 15, 16, 17) | transition='15' | top feature: alt_util_ln Progress_t13
value > 0 → higher util raises P(chose '15'); value < 0 → lowers it; value ≈ 0 → no effect
alt_util_ln Progress_t13 <= 0.8816
samples = 819
value = 0.0001
True
False
alt_util_ln Progress_t13 <= 0.8658
samples = 607
value = 0.0
hour <= 11.5
samples = 212
value = 0.0005
hour <= 17.5
samples = 499
value = -0.0
hour <= 13.5
samples = 108
value = 0.0001
hour <= 8.5
samples = 104
value = 0.0001
alt_util_ln Progress_t13 <= 0.8974
samples = 108
value = 0.0008
samples = 469
value = -0.0
samples = 30
value = -0.0
samples = 68
value = 0.0
samples = 40
value = 0.0002
samples = 46
value = -0.0
samples = 58
value = 0.0002
samples = 49
value = 0.0005
samples = 59
value = 0.0011


CATE — choice_set=(13, 14, 15, 16, 17) | transition='16' | n=819 | ATE=-0.1895 std=0.8699
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=-0.1895
Count
CATE θ(X)
Mean CATE by resource
Vesa
Tomas
Timothy
Srinivasan
Samira
Saki
Roland
Robert
Rikard
Rakesh
Praveen
Peter
Per
Niklas
Mikael
Martin
Maria
Marc
Leif
Lars
Katarina
Jonas
Ing-Marie
Jan
Hineesh
Ewa
David
Christer
Britt
Ann-Charlotte
Alan
Mean CATE


CATE summary tree | choice_set=(13, 14, 15, 16, 17) | transition='16' | top feature: alt_util_ln Progress_t13
value > 0 → higher util raises P(chose '16'); value < 0 → lowers it; value ≈ 0 → no effect
alt_util_ln Progress_t13 <= 0.8921
samples = 819
value = -0.1895
True
False
hour <= 15.5
samples = 650
value = -0.4496
dayofweek <= 2.5
samples = 169
value = 0.8107
dayofweek <= 1.5
samples = 568
value = -0.583
dayofweek <= 2.5
samples = 82
value = 0.4744
alt_util_ln Progress_t13 <= 0.9079
samples = 95
value = 0.2651
alt_util_ln Progress_t13 <= 0.9026
samples = 74
value = 1.5112
samples = 210
value = -0.2254
samples = 358
value = -0.7928
samples = 49
value = 0.7258
samples = 33
value = 0.101
samples = 57
value = -0.0372
samples = 38
value = 0.7185
samples = 34
value = 0.9667
samples = 40
value = 1.974


CATE — choice_set=(13, 14, 15, 16, 17) | transition='17' | n=819 | ATE=-0.3002 std=0.4596
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=-0.3002
Count
CATE θ(X)
Mean CATE by resource
Vesa
Tomas
Timothy
Srinivasan
Samira
Saki
Roland
Robert
Rikard
Rakesh
Praveen
Peter
Per
Niklas
Mikael
Martin
Maria
Marc
Leif
Lars
Katarina
Jonas
Ing-Marie
Ian
Hineesh
Ewa
David
Christer
Britt
Ann-Charlotte
Alan
Mean CATE


CATE summary tree | choice_set=(13, 14, 15, 16, 17) | transition='17' | top feature: hour
value > 0 → higher util raises P(chose '17'); value < 0 → lowers it; value ≈ 0 → no effect
hour <= 9.5
samples = 819
value = -0.3002
True
False
alt_util_ln Progress_t13 <= 0.0947
samples = 255
value = -0.8195
dayofweek <= 0.5
samples = 564
value = -0.0655
hour <= 7.5
samples = 67
value = -1.156
hour <= 7.5
samples = 188
value = -0.6996
alt_util_ln Progress_t13 <= 0.8342
samples = 98
value = -0.3528
alt_util_ln Progress_t13 <= 0.8868
samples = 466
value = -0.005
samples = 27
value = -1.4501
samples = 40
value = -0.9574
samples = 64
value = -0.9259
samples = 124
value = -0.5828
samples = 31
value = -0.0634
samples = 67
value = -0.4867
samples = 376
value = -0.0694
samples = 90
value = 0.2641

CATE statistics summary
cate_std = heterogeneity; pct_positive/negative = share of cases where util helps/hurts; top_feature = main moderator

choice_set	transition	n	ate	cate_std	cate_min	cate_p25	cate_p50	cate_p75	cate_max	pct_positive	pct_negative	top_feature
(0, 1, 2, 3, 4, 12)	0	949	-5.828992	2.043733	-10.949063	-7.145732	-6.007822	-4.65788	-0.535804	0.0	100.0	alt_util_Closed_t2
(0, 1, 2, 3, 4, 12)	1	949	-5.295144	1.542446	-9.308237	-6.270431	-5.268665	-4.139984	-1.582065	0.0	100.0	alt_util_Awaiting Assignment_t3
(0, 1, 2, 3, 4, 12)	2	949	-4.256705	3.400557	-13.309517	-6.538341	-4.106236	-1.785116	3.279951	10.7	89.3	alt_util_Awaiting Assignment_t3
(0, 1, 2, 3, 4, 12)	3	949	-2.457923	2.542776	-8.623436	-3.82317	-2.40139	-1.337512	3.355329	16.6	83.4	alt_util_Closed_t2
(0, 1, 2, 3, 4, 12)	4	949	-0.476254	1.198376	-2.975692	-1.317841	-0.730209	0.131542	3.119681	28.1	71.9	alt_util_Closed_t2
(5, 6, 7, 8, 9, 12)	5	134	-2.088639	0.755998	-3.140853	-2.877732	-2.29407	-1.285561	-0.913971	0.0	100.0	alt_util_Awaiting Assignment_t8
(5, 6, 7, 8, 9, 12)	6	134	0.423645	0.217579	0.107389	0.246214	0.391036	0.558046	0.90764	100.0	0.0	alt_util_Closed_t7
(5, 6, 7, 8, 9, 12)	7	134	-1.960342	0.686216	-3.481121	-2.420461	-1.779389	-1.352981	-1.007661	0.0	100.0	alt_util_In Progress_t5
(5, 6, 7, 8, 9, 12)	8	134	-1.046883	0.159861	-1.44912	-1.160739	-1.03584	-0.933816	-0.703616	0.0	100.0	alt_util_Closed_t7
(5, 6, 7, 8, 9, 12)	9	134	-0.263324	0.195299	-0.587505	-0.458433	-0.173452	-0.089055	-0.047466	0.0	100.0	alt_util_Closed_t7
(10, 11, 12)	10	273	-0.415249	0.331752	-1.213245	-0.660435	-0.398244	-0.162429	0.236246	11.7	88.3	hour
(10, 11, 12)	11	273	-0.234641	0.456303	-1.217657	-0.6228	-0.247957	0.098713	0.628433	36.6	63.4	alt_util_In Progress_t10
(13, 14, 15, 16, 17)	13	819	-0.133493	0.928143	-3.119483	-0.696402	0.021269	0.508202	1.988545	51.5	48.5	dayofweek
(13, 14, 15, 16, 17)	14	819	-0.308369	0.235001	-1.819752	-0.420865	-0.270347	-0.151781	0.133446	2.0	98.0	dayofweek
(13, 14, 15, 16, 17)	15	819	0.000131	0.000371	-0.000418	-8e-06	2e-06	6.5e-05	0.002802	56.2	43.8	alt_util_In Progress_t13
(13, 14, 15, 16, 17)	16	819	-0.189537	0.869862	-2.42497	-0.801417	-0.256895	0.294288	2.713001	36.5	63.5	alt_util_In Progress_t13
(13, 14, 15, 16, 17)	17	819	-0.300239	0.459564	-1.736501	-0.583136	-0.169229	-0.001601	0.809797	24.8	75.2	hour

Key Assumptions & Interpretation Notes

▸ Backdoor criterion (all methods)

X = {hour, dayofweek, n_alternatives, res_*} must block all confounding paths T ← C → Y.

Any variable that affects both util and routing must be in X. Unobserved confounders

cannot be detected and will bias all estimates.

▸ SUTVA (all methods)

One case's routing does not affect another's treatment or outcome.

Violated if resource occupation by one case changes the util seen by a concurrent case.

▸ Positivity (all methods)

Every covariate combination X must have positive probability for all util levels T.

Diagnosed by the covariate balance (SMD) in the backdoor check.

▸ Linearity (Double ML only)

θ is the best-fit linear slope. If the effect varies with X, θ is a variance-weighted

average of CATEs — valid as a summary, but not the simple mean.

Double ML is more powerful than the forest for testing 'is there any effect?'

▸ Honest splitting (Causal Forest only)

Tree structure and leaf estimates use separate subsamples. Yields valid CIs but

halves effective sample size → forest CIs are wider than Double ML's.

A significant Double ML θ with an insignificant forest ATE is not a contradiction.

▸ What the analysis cannot establish

Causal direction (assumes util → choice, not the reverse).

External validity beyond the observed covariate distribution.

Dynamic/lagged effects across cases over time.

Full documentation: ASSUMPTIONS.md